Arrow of Time and Complexification from Self-Modeling Thermodynamics

Bryan Ehrlich

A companion paper derived the Standard Model gauge group and its chiral fermion representation from the exceptional Jordan algebra $h_3(\mathbb{O})$, conditional on complexifying the real Peirce half-space $V_{1/2} \cong \mathbb{O}^2$ to $V_{1/2}^{\mathbb{C}} = S_{10}^+$—a step left open (Gap C) after four algebraic routes were shown to fail. This paper proves two independent results. First, the Landauer bound on the self-modeling update cycle, together with an entropy gradient theorem (three independent routes), shows that every self-modeler requires a non-equilibrium entropy gradient and hence a time-oriented spacetime. Second, a three-consequence theorem shows that the single algebraic choice $u \in S^6$ fixing the gauge group and chirality also requires spacetime time-orientation for physical Weyl spinors, since Γ_* contains the timelike generator Γ_0. Both chains converge on time-orientation as a shared geometric prerequisite—connecting particle physics to observer thermodynamics. This narrows Gap C: complexification is necessary for Standard Model–like observers; whether non-complexified blocks support non-SM self-modelers remains open. The Landauer entropy deficit ($\sim 10^{28}$ nats) is 94 orders below Penrose's gravitational estimate ($\sim 10^{122}$ nats); the exhaustion timescale exceeds the universe age by $\sim 10^{93}$. The bridge from observer thermodynamics to particle physics rests not on logical closure but on structural convergence: two independent arguments identify the same geometric structure as indispensable.

I. INTRODUCTION

Why does particle physics use complex numbers? The Standard Model is a quantum field theory over $\mathbb{C}$: its gauge group $\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ acts on complex representations, and its chiral fermions are complex Weyl spinors. The exceptional Jordan algebra $h_3(\mathbb{O})$—the 3×3 Hermitian matrices over the octonions $\mathbb{O}$—provides a natural algebraic setting in which the Standard Model gauge group and its chiral representation emerge [1–5]. In a companion paper [6], we showed that a single algebraic choice—a unit imaginary octonion $u \in S^6 \subset \mathrm{Im}(\mathbb{O})$—simultaneously determines both the gauge group $\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ (via the F_4 intersection of Todorov–Drenska [2]) and the chiral (left-handed) fermion representation (via the Cl(6) volume form of Furey [3]). This "one choice, two consequences" result connects the self-modeling axioms of Paper 5 [7] through $h_3(\mathbb{O})$ to a one-generation chiral Standard Model representation in a nine-link chain with explicit gap identification.

The derivation, however, is conditional on a step that Paper 7 left open. The Peirce decomposition of $h_3(\mathbb{O})$ under a rank-1 idempotent E_{11} produces a half-space $V_{1/2} \cong \mathbb{O}^2$ that is a *real* 16-dimensional space carrying the spinor representation of Spin(9). The gauge group and chirality results require this space to be complexified: $V_{1/2} \otimes_{\mathbb{R}} \mathbb{C} = S_{10}^+$, upgrading $\mathrm{Spin}(9) \to \mathrm{Spin}(10)$ and $F_4 \to E_6$. Paper 7 argued that the C*-algebra nature of the observer (from Paper 5 [7]) motivates this complexification, but flagged it as an open gap: **Gap C**.

A. The problem: complexification is not algebraically forced

Paper 7's gap analysis [6] showed that four independent algebraic routes to complexification all fail. The obstruction is structural: the Peirce eigenspace $V_1 = \mathbb{R} \cdot E_{11}$ is one-dimensional, creating a bottleneck through which no Peirce-mediated mechanism can transmit a complex structure from the observer's C*-algebra to the half-space $V_{1/2}$. This result is sharp. Complexification of $V_{1/2}$ is *not* forced by the Jordan algebra structure of $h_3(\mathbb{O})$, by the Peirce decomposition, or by any known algebraic property of the observer–universe interface [5].

The failure is not a computational gap awaiting a cleverer argument. It is a no-go result: the V_1 bottleneck is a theorem about the Peirce decomposition, and it closes the algebraic route to Gap C. Any resolution must come from outside the algebra.

B. The resolution: thermodynamic selection

We narrow Gap C via two independent chains of results that converge on a common geometric prerequisite.

Chain 1 (all self-modelers): The Landauer bound [8, 9] applied to the self-modeling update cycle (Sec. III) shows that maintaining correlations $I(B;M) > 0$ dissipates at least $k_{\mathrm{B}} T\, I(B;M)$ per cycle. Free energy requires non-equilibrium ($F > F_{\mathrm{eq}}$), and a finite system out of equilibrium must have $S(t) < S_{\max}$. The entropy gradient theorem (Sec. V A), proved via three independent routes, establishes this for *all* self-modelers regardless of their matter content. The entropy gradient defines a temporal direction—the arrow of time.

Chain 2 (SM-like observers): Chirality requires complexification: the Cl(6) volume form ω_6 has no eigenstates in real $V_{1/2} = \mathbb{O}^2$ [6]. Chirality also requires time-orientation: the chirality operator Γ_* in Lorentzian signature flips under time reversal [10] (Sec. IV). Therefore SM-like observers require both complexification and time-orientation.

Structural convergence: Both chains require time-orientation—Chain 1 for the thermodynamic arrow, Chain 2 for the Weyl decomposition. The single algebraic choice $u \in S^6$ that determines gauge group and chirality [6] simultaneously creates a requirement for the same time-orientation that the entropy gradient provides. This convergence is a structural consistency check—the framework's particle physics and thermodynamics both require a geometric property that the emergent spacetime provides. It is *not* a logical closure: the two chains are independent, and we do not prove that non-complexified blocks have $\rho = 0$.

Epistemic honesty. Gap C is *narrowed*, not *closed*. "Narrowed" means we prove complexification is necessary for SM-like observers, but do not prove $\rho = 0$ for all non-complexified blocks. A non-complexified block could in principle have time-orientation (from dynamics), an entropy gradient, and non-SM self-modelers with vector-like rather than chiral matter. Whether such physics supports self-modeling is open. This does not undermine the program: it explains *our* physics (we observe SM chirality, which requires complexification), not all possible physics—the same epistemic status as "why 3+1 dimensions?"

C. What this paper proves and does not prove

Proved (within stated assumptions):

- **Entropy gradient theorem** (Theorem 10): every self-modeling composite with $\rho > 0$ satisfies $S(t) < S_{\max}$, established via three independent routes using different assumption subsets (Sec. V A).
- **Landauer bound on self-modeling** (Theorem 2): each update cycle dissipates at least $k_{\mathrm{B}} T\, I(B; M)$ of work (Sec. III B).
- **Three-consequence theorem** (Theorem 7): the single choice $u \in S^6$ determines the gauge group, chirality, *and* the requirement for spacetime time-orientation—extending Paper 7's "one choice, two consequences" to three (Sec. IV E).
- **Gap C narrowing** (Theorem 11): complexification is necessary for SM-like observers; the status of non-SM self-modelers in non-complexified blocks remains open (Sec. V C).

Not proved:

- **Gap C is narrowed, not closed.** The V_1 bottleneck remains genuine [5]. Non-complexified blocks exist and are not forbidden by the algebra. Whether they can support non-SM self-modelers is an open question.
- **The Past Hypothesis is not derived.** The entropy gradient theorem shows that self-modeling *requires* $S(t) < S_{\max}$ but does not explain *why* the initial entropy was low [11, 12]. Its status is elevated from "observed cosmological fact" to "necessary condition for self-modeling"—an observer-selection argument, not a first-principles derivation.
- **The Second Law is not derived.** It is *used* as input (standard physics) in Link 2 of the chain.
- **The spectral action is not computed.** The dynamics (GR + SM Lagrangian from the Connes–Chamseddine–Marcolli spectral triple [13]) remains a separate open problem.

D. Paper organization

Section II derives the entropy dynamics of self-modeling composites under the SWAP Hamiltonian: the CPTP channel, its unitality, and three regimes of entropy behavior (oscillation, monotonic increase, effective thermalization).

Section III establishes the Landauer bound on self-modeling. The self-modeling update cycle maps onto Bennett's Maxwell's demon analysis [9], and the Reeb–Wolf quantum Landauer bound [14] gives the minimum dissipation per cycle. Three independent arguments close a potential coherence loophole.

Section IV shows that physical realization of the chirality from Paper 7 requires the emergent spacetime to carry a time-orientation, extending "one choice, two consequences" to "one choice, three consequences."

Section V brings the preceding results together. The entropy gradient theorem is proved via three independent routes. Two independent chains—one thermodynamic, one algebraic—converge on time-orientation, narrowing Gap C. The master theorem collects the results, and an assumption register classifies every input by severity.

Section VI extracts quantitative predictions: the minimum entropy deficit for self-modeling, comparison with Penrose's gravitational entropy estimate [11] (a 94-order-of-magnitude gap that is a feature, not a failure), the exhaustion timescale, and the experiential density profile.

Section VII places the results in the context of the self-modeling program (Papers 5–7) and discusses connections to the spectral triple approach [13], Jacobson's thermodynamic gravity [15], and observer-selection principles [16].

Appendix A collects detailed derivations, and Appendix B provides the complete assumption register.

II. ENTROPY UNDER SEQUENTIAL PRODUCT DYNAMICS

The thermodynamic arrow of time for a self-modeler is not postulated but derived from the quantum channel induced by the SWAP Hamiltonian on body–model composites [7, 17]. In this section we establish the entropy

dynamics that underlies the Landauer bound of Sec. III and the complexification gradient of Sec. V. The central results are: (i) an explicit CPTP channel for SWAP dynamics when the model is maximally mixed, (ii) a proof that this channel is *unital*, so that Lindblad's entropy inequality [18] applies, and (iii) a classification of entropy behavior into three physically distinct regimes.

A. SWAP channel on a qubit pair

Consider a body B and model M, each a qubit ($\mathcal{H}_B = \mathcal{H}_M = \mathbb{C}^2$). The SWAP Hamiltonian derived in Ref. [17] from diagonal $\mathrm{U}(n)$ covariance is

$$H = J\,F\,, \tag{1}$$

where F is the SWAP operator ($F|ij\rangle = |ji\rangle$, $F^2 = \mathbb{1}$) and $J > 0$ is the coupling strength. Because $F^2 = \mathbb{1}$, the unitary evolution takes the closed form

$$U(t) = e^{-iJFt} = \cos(Jt)\,\mathbb{1} - i\sin(Jt)\,F\,. \tag{2}$$

For a product initial state $\rho_B \otimes \rho_M$ with the model in the maximally mixed state $\rho_M = \mathbb{1}/2$ (maximal prior ignorance), the evolved reduced state of B is obtained by tracing over M. A direct computation using $F(\rho_B \otimes \rho_M)F = \rho_M \otimes \rho_B$ and $[\mathbb{1}/2, \rho_B] = 0$ yields

$$\mathcal{E}(\rho_B) = \cos^2(Jt)\;\rho_B + \sin^2(Jt)\;\frac{\mathbb{1}}{2}\,. \tag{3}$$

This is a *depolarizing channel* with parameter $p = \sin^2(Jt)$:

$$\mathcal{E}(\rho_B) = (1-p)\,\rho_B + p\,\frac{\mathbb{1}}{2}\,, \qquad p = \sin^2(Jt) \in [0,1]\,. \tag{4}$$

The channel admits the Kraus representation $\mathcal{E}(\rho) = \sum_{i=0}^{3} K_i \rho\, K_i^\dagger$ with

$$K_0 = \sqrt{1 - \tfrac{3p}{4}}\;\mathbb{1}\,, \qquad K_j = \frac{\sqrt{p}}{2}\,\sigma_j \quad (j = 1,2,3)\,, \tag{5}$$

where σ_j are the Pauli matrices. One verifies $\sum_i K_i^\dagger K_i = \mathbb{1}$ (trace preservation) by inspection.

a. Unitality. A CPTP map is *unital* if $\mathcal{E}(\mathbb{1}/d) = \mathbb{1}/d$. For the channel (3),

$$\mathcal{E}\left(\frac{\mathbb{1}}{2}\right) = \cos^2(Jt)\;\frac{\mathbb{1}}{2} + \sin^2(Jt)\;\frac{\mathbb{1}}{2} = \frac{\mathbb{1}}{2}\,. \tag{6}$$

The channel is therefore unital for all t and all J. This is proved, not assumed: the forbidden proxy of asserting unitality without demonstration does not arise.

b. General model state. For an arbitrary model state $\rho_M \neq \mathbb{1}/2$, the cross terms involving $[\rho_M, \rho_B]$ no longer vanish, and the reduced dynamics becomes

$$\rho_B(t) = \cos^2(Jt)\;\rho_B + \sin^2(Jt)\;\rho_M - i\sin(Jt)\cos(Jt)\;[\rho_M, \rho_B]\,. \tag{7}$$

In this case $\mathcal{E}(\mathbb{1}/2) = \cos^2(Jt)\,(\mathbb{1}/2) + \sin^2(Jt)\,\rho_M \neq \mathbb{1}/2$, so the channel is *not* unital unless $\rho_M = \mathbb{1}/2$. We return to the physical significance of this distinction below; for now we note that entropy can decrease for non-maximally-mixed model states (verified numerically: 56% of state–time points show $\Delta S < 0$ for $\rho_M = \mathrm{diag}(0.9, 0.1)$).

B. Entropy change per interaction step

We now quantify the entropy change when $\rho_M = \mathbb{1}/2$. Write $\rho_B = \mathrm{diag}(\lambda, 1-\lambda)$ in its eigenbasis, with $\lambda \in [0,1]$ the larger eigenvalue. After one application of (4), the eigenvalue shifts to

$$\lambda' = \lambda + p\big(\tfrac{1}{2} - \lambda\big) = (1-p)\,\lambda + \frac{p}{2}\,, \tag{8}$$

which is a convex combination of λ and $1/2$, hence $|\lambda' - 1/2| \leq |\lambda - 1/2|$ for all $p \in (0,1]$. The von Neumann entropy is $S_{\mathrm{vN}}(\rho_B) = h(\lambda)$, where

$$h(x) = -x\ln x - (1-x)\ln(1-x) \tag{9}$$

is the binary entropy in nats. The single-step entropy change is therefore

$$\Delta S = h\big(\lambda + p(\tfrac{1}{2} - \lambda)\big) - h(\lambda) \geq 0\,, \tag{10}$$

with equality if and only if $\lambda = 1/2$ (already maximally mixed) or $p = 0$ (no interaction). The inequality follows from the strict concavity of h and the fact that λ' is closer to the maximum at $1/2$ than λ. Equivalently, it follows from Lindblad's theorem [18]: a doubly stochastic (trace-preserving and unital) quantum channel cannot decrease the von Neumann entropy.

C. Three regimes of entropy dynamics

The single-step inequality (10) does not by itself guarantee monotonic entropy increase over time. The behavior depends critically on the dynamical context. We distinguish three physically distinct regimes.

Proposition 1 (Entropy regimes under SWAP dynamics)**.** *Let $H = JF$ act on a system of N qubits. Let $\rho_1(t) = \mathrm{Tr}_{2,\ldots,N}[\rho(t)]$ denote the reduced state of site 1, with $\rho(t) = U(t)\,\rho(0)\,U(t)^\dagger$. Then:*

Regime I *(closed 2-site system, $N = 2$): $S_{\mathrm{vN}}(\rho_1(t))$ is periodic with period $T_S = \pi/J$. Entropy oscillates and does not increase monotonically.*

Regime II *(repeated interaction with fresh bath): If at each step $k = 1, 2, \ldots$ the body interacts with a fresh model qubit M_k prepared in $\mathbb{1}/2$, then $S_{\mathrm{vN}}(\mathcal{E}^N(\rho_B))$ is monotonically non-decreasing in N, converging geometrically to $\ln 2$.*

Regime III *(N-site lattice, $N \gg 1$): $S_{\mathrm{vN}}(\rho_1(t))$ increases toward $\ln 2$ on a relaxation timescale $t_{\mathrm{relax}} \sim 1/J$,*

with fluctuations suppressed as N grows. Poincaré recurrence occurs only at $t_{\rm rec} \sim e^{c\cdot 2^N}$, which is physically irrelevant for macroscopic systems.

We now establish each regime in turn.

a. Regime I: Oscillation. For the closed 2-site system, the composite state evolves unitarily and $\rho_1(t)$ is given exactly by Eq. (3). Since $\cos^2(Jt)$ and $\sin^2(Jt)$ have period π/J,

$$\rho_1(t+\pi/J) = \rho_1(t) \qquad \Longrightarrow \qquad S(\rho_1(t+\pi/J)) = S(\rho_1(t))\,. \tag{11}$$

For an initial pure state $\rho_B = |0\rangle\langle 0|$ with $\rho_M = \mathbb{1}/2$, the entropy rises from 0 to $\ln 2$ (at $Jt = \pi/2$) and returns to 0 (at $Jt = \pi$) in each cycle. This is a direct counterexample to naive monotonicity on small systems and provides an important internal consistency check: the formalism does not claim more than can be derived.

Note that Eq. (3) describes the exact closed-system dynamics, not iterated application of the channel with fresh baths. In the 2-site case, ρ_M becomes entangled with ρ_B after the first instant, so re-applying the single-step channel (4) would be incorrect.

b. Regime II: Monotonic increase. In the repeated interaction model, the body interacts with a succession of independent model qubits, each prepared in $\mathbb{1}/2$. The channel (4) is applied *de novo* at each step with a fresh, uncorrelated bath. After N steps, an induction argument gives the iterated channel

$$\mathcal{E}^N(\rho_B) = (1-p)^N\,\rho_B + \big(1-(1-p)^N\big)\,\frac{\mathbb{1}}{2}\,, \tag{12}$$

which is itself a depolarizing channel with effective parameter $p_N = 1-(1-p)^N$. The eigenvalue after N steps is

$$\lambda_N = (1-p)^N\,\lambda_0 + \frac{1-(1-p)^N}{2}\,, \tag{13}$$

converging geometrically to $1/2$. Since each step applies a unital CPTP map, Lindblad's theorem [18] gives the monotonicity chain

$$S\big(\mathcal{E}^{N+1}(\rho)\big) \geq S\big(\mathcal{E}^N(\rho)\big) \geq \cdots \geq S\big(\mathcal{E}(\rho)\big) \geq S(\rho)\,. \tag{14}$$

The rate of approach to maximum entropy follows from expanding $h(\lambda)$ about $\lambda = 1/2$: $h(\lambda) \approx \ln 2 - 2(\lambda - 1/2)^2 + \mathcal{O}\big((\lambda-1/2)^4\big)$. Substituting $\lambda_N - 1/2 = (1-p)^N(\lambda_0 - 1/2)$ yields

$$\ln 2 - S_N \approx 2\,(1-p)^{2N}\,(\lambda_0 - \tfrac{1}{2})^2\,, \tag{15}$$

so the entropy deficit decays geometrically with base $(1-p)^2 < 1$.

c. Regime III: Effective thermalization. On an N-site chain with Hamiltonian $H = J\sum_{i=1}^{N-1} F_{i,i+1}$, the dynamics is unitary on the full Hilbert space ($\dim = 2^N$). For any single site, the remaining $N-1$ sites act as an effective bath. As N increases, the entropy of a single-site reduced state evolves as follows:

1. It increases from its initial value on a timescale $t_{\rm relax} \sim 1/J$.
2. It fluctuates near the equilibrium value $\ln 2$ with fluctuations that decrease with system size.
3. It returns to its initial value only at the Poincaré recurrence time $t_{\rm rec} \sim e^{c\cdot 2^N}$.

Numerical simulations of lattice sizes $N = 2, 4, 6$ confirm the suppression of fluctuations: the standard deviation of the late-time entropy is 0.246 ($N = 2$), 0.132 ($N = 4$), and 0.004 ($N = 6$), demonstrating rapid approach to effectively irreversible behavior. For any macroscopic self-modeler ($N \gg 1$), the recurrence time exceeds the age of the universe by an exponentially large factor, and the entropy increase is as robust as the Second Law of thermodynamics.

D. Non-unital concern and its resolution

A potential objection is that the Lüders product underlying the sequential dynamics [7] is generically non-unital for individual measurement outcomes. This concern is resolved by distinguishing the relevant dynamical scenarios:

1. **The physically relevant channel is unital.** Equation (6) proves that the reduced channel for $\rho_M = \mathbb{1}/2$ is unital. This is not an assumption but a consequence of the SWAP symmetry: $F(\rho_B \otimes \mathbb{1}/2)F = (\mathbb{1}/2) \otimes \rho_B$, so the partial trace yields $\mathbb{1}/2$ regardless of ρ_B.
2. **Maximally mixed model is the correct starting point.** Before any observation, the model's state encodes maximal ignorance about the body, corresponding to $\rho_M = \mathbb{1}/2$. In the repeated interaction framework (Regime II), each model qubit is prepared fresh in this state.
3. **The non-unital case is documented, not hidden.** For $\rho_M \neq \mathbb{1}/2$, the channel (7) is non-unital and entropy can decrease. This is a genuine physical effect—a biased model can extract information from the body—not a gap in the argument. The arrow of time follows from $\rho_M = \mathbb{1}/2$, and this starting condition is itself motivated by the self-modeling framework.

E. Summary and connection to the Landauer bound

The results of this section establish that self-modeling composites governed by the SWAP Hamiltonian (1) experience monotonic entropy increase under the conditions relevant to macroscopic self-modelers: either repeated interaction with fresh maximally mixed bath qubits (exact

monotonicity, Eq. (14)) or embedding in a large lattice (effective monotonicity for $t \ll t_{\rm rec}$). The entropy converges geometrically to $S_{\max} = \ln 2$ per qubit, with the rate set by the depolarizing parameter $p = \sin^2(Jt)$.

This entropy increase is not a postulate but a *theorem* for the repeated interaction model and a standard statistical-mechanical consequence for lattice systems. In the next section, we show that this entropy production implies a minimum thermodynamic cost for self-modeling, via the Landauer bound.

III. SELF-MODELING REQUIRES FREE ENERGY

The entropy dynamics of Sec. II established that SWAP interactions drive the body toward thermal equilibrium, erasing the body–model correlations that define self-modeling. We now show that maintaining these correlations has an irreducible thermodynamic cost: every self-modeling update cycle must dissipate at least $k_{\rm B}T\, I(B;M)$ of free energy, where $I(B;M)$ is the mutual information between body and model. This is the Landauer bound [8] applied to the erasure step that every self-modeler must perform.

The section is organized as follows. We first identify the self-modeling cycle as an instance of Bennett's Maxwell's demon analysis [9] (Sec. III A). We then derive the Landauer bound in the quantum setting using the Reeb–Wolf result [14] (Sec. III B). We close a potential coherence loophole via three independent arguments (Sec. III C), and finally assemble the three-link chain connecting self-modeling to the need for an entropy gradient (Sec. III D).

A. The self-modeler as Maxwell's demon

A self-modeling composite $(B, M, \&)$ in the sense of Ref. [7] consists of a body B (Hilbert space $\mathcal{H}_B$, dim $= d_B$), a model M ($\mathcal{H}_M$, dim $= d_M$), and the Lüders sequential product $a \,\&\, b = \sqrt{a}\, b \sqrt{a}$ implementing updates. Faithful self-modeling requires $I(B;M) > 0$: the model must carry non-trivial information about the body.

Each update cycle decomposes into three information-processing steps [9]:

1. **Test.** Interact B and M to acquire information about the current state of B.

2. **Erase.** Clear the old tracking data from M. This step is logically irreversible: the $I(B;M)$ nats of correlation between B and M are destroyed.

3. **Write.** Record fresh tracking data into M, restoring $I(B;M) > 0$.

The writing step can in principle be performed reversibly, adding no thermodynamic cost beyond that of erasure [9]. The erasure step cannot: it maps many possible states of M to a single reset state, an operation that is logically irreversible. This is precisely the structure of Maxwell's demon, and the thermodynamic cost of the erasure step is bounded below by Landauer's principle [8].

We emphasize that this decomposition is not an approximation. The Lüders product is a quantum operation (completely positive map), and the non-selective version $\Lambda(\rho) = \sqrt{a}\,\rho\,\sqrt{a} + \sqrt{\mathbb{1} - a}\,\rho\,\sqrt{\mathbb{1} - a}$ is CPTP, as established in Sec. II A. The Bennett decomposition applies to any information-processing cycle that maintains correlations and periodically refreshes them [9].

B. Landauer bound on the self-modeling cycle

The mutual information between body and model,

$$I(B;M) = S_{\rm vN}(\rho_B) + S_{\rm vN}(\rho_M) - S_{\rm vN}(\rho_{BM})\,, \qquad (16)$$

quantifies the information content that the erasure step must destroy. Here $S_{\rm vN}(\rho) = -\operatorname{Tr}(\rho \ln \rho)$ is the von Neumann entropy in nats.

The quantum Landauer bound [14] states that resetting a quantum register with entropy $S_{\rm vN}(\rho_S)$, in thermal contact with a bath at temperature T, requires work $W \geq k_{\rm B}T\, S_{\rm vN}(\rho_S)$. Applied to the erasure of the old body–model correlation:

Theorem 2 (Landauer bound on self-modeling)**.** *Let $(B, M, \&)$ be a self-modeling composite with finite-dimensional Hilbert spaces, Lüders sequential product, and faithful tracking $I(B;M) > 0$. Suppose the composite is in thermal contact with a heat bath at temperature T. Then each self-modeling update cycle dissipates work*

$$\boxed{W_{\min} = k_{\rm B}T\, I(B;M)} \qquad (17)$$

with equality for quasi-static (reversible) erasure protocols.

Proof. The update cycle erases $I(B;M)$ nats of correlation (step 2 of the Bennett decomposition). By the quantum Landauer bound [14], this requires $W_{\rm erase} \geq k_{\rm B}T\, I(B;M)$. The writing step (step 3) can be performed reversibly [9], contributing zero net work. Hence $W_{\rm cycle} \geq k_{\rm B}T\, I(B;M)$. □

Assumptions.—The theorem requires:

A1. Finite-dimensional quantum mechanics (standard).

A2. Thermal contact with a bath at temperature T (standard Landauer regime).

a. Experiential density and its cost. The experiential density from Ref. [7],

$$\rho = I(B;M)\left(1 - \frac{I(B;M)}{S_{\rm vN}(\rho_B)}\right), \qquad (18)$$

is a concave function of $I(B;M)$, peaking at $I^* = S_{\rm vN}(\rho_B)/2$ with $\rho_{\max} = S_{\rm vN}(\rho_B)/4$. The Landauer cost at peak experiential density is

$$W_{\rm peak} \geq k_{\rm B}T\,\frac{S_{\rm vN}(\rho_B)}{2}\,. \tag{19}$$

For a qubit body ($d_B = 2$, $S_{\rm vN}(\rho_B) \leq \ln 2$), this gives $W_{\rm peak} \geq k_{\rm B}T\,(\ln 2)/2 \approx 0.35\,k_{\rm B}T$.

b. Equilibrium death. At thermal equilibrium, $\rho_{BM} = \mathbb{1}/(d_B d_M)$. The marginals are $\rho_B = \mathbb{1}/d_B$ and $\rho_M = \mathbb{1}/d_M$, so

$$I(B;M)^{\rm eq} = \ln d_B + \ln d_M - \ln(d_B d_M) = 0\,. \tag{20}$$

The body and model are uncorrelated: the model contains no information about the body. From Eq. (18), $\rho^{\rm eq} = 0$. Self-modeling at thermal equilibrium is impossible—there is nothing to model and no free energy with which to model it.

c. Free energy depletion rate. A self-modeler cycling at frequency ν depletes free energy at a rate bounded below by

$$\frac{dF}{dt} \leq -k_{\rm B}T\,I(B;M)\,\nu\,. \tag{21}$$

C. Closing the coherence loophole

A potential objection is that quantum coherence, as a resource absent in classical thermodynamics, might allow a self-modeler to maintain $I(B;M) > 0$ without paying the full Landauer cost. We close this loophole with three independent arguments.

a. R1: The Lüders product destroys coherence (CPTP). Let the effect a have spectral decomposition $a = \sum_i \lambda_i |i\rangle\langle i|$. The non-selective Lüders channel acts on off-diagonal matrix elements as

$$\Lambda(\rho)_{ij} = c_{ij}\,\rho_{ij}\,, \qquad c_{ij} = \sqrt{\lambda_i\lambda_j} + \sqrt{(1-\lambda_i)(1-\lambda_j)}\,, \tag{22}$$

while preserving the diagonal ($c_{ii} = 1$). By the Cauchy–Schwarz inequality applied to the vectors $(\sqrt{\lambda_i},\,\sqrt{1-\lambda_i})$ and $(\sqrt{\lambda_j},\,\sqrt{1-\lambda_j})$,

$$c_{ij} \leq 1 \tag{23}$$

with equality if and only if $\lambda_i = \lambda_j$. Therefore, whenever the effect a has distinct eigenvalues, the self-modeling update strictly reduces off-diagonal coherence in the eigenbasis of a. This blocks evasion strategies that rely on unitary-only protocols (the Lüders product is mandatory in the Paper 5 framework [7]) or on storing information in quantum coherences (repeated Lüders updates drive all off-diagonals to zero).

b. R2: Maintaining coherence costs free energy. Even setting aside the Lüders product, thermal contact with a bath at temperature T induces decoherence. The Gibbs state $\rho^{\rm eq} = e^{-\beta H}/Z$ is diagonal in the energy eigenbasis; all off-diagonal coherences vanish at equilibrium. Open-system dynamics drives any initial coherence to zero at a rate $\Gamma_d = (\gamma_+ + \gamma_-)/2 > 0$, where $\gamma_\pm$ are the bath emission and absorption rates. Maintaining coherence against this decay requires active intervention—error correction or continuous driving—which itself costs free energy. This argument is framework-independent: it does not rely on the Lüders product or the specific axioms of Paper 5.

c. R3: Sagawa–Ueda cross-check. The generalized Jarzynski equality for feedback-controlled systems [19],

$$\left\langle e^{-\beta(W-\Delta F)} \right\rangle = e^{I_{fc}}\,, \tag{24}$$

where I_{fc} is the efficacy of feedback control (transfer entropy), independently confirms that any measurement-feedback cycle has a minimum thermodynamic cost. By Jensen's inequality, the mean work satisfies $\langle W\rangle \geq \Delta F - k_{\rm B}T\,I_{fc}$. For a cyclic process ($\Delta F = 0$), $\langle W\rangle \geq -k_{\rm B}T\,I_{fc}$, so the maximum *extractable* work is $k_{\rm B}T\,I_{fc}$. Since the self-modeling cycle maps onto the Sagawa–Ueda framework (body → system, model → memory, Lüders product → measurement, update → feedback), the bound is consistent with Theorem 2. The detailed mapping and numerical consistency check (50 random states, all satisfying both bounds) are reported in the supplemental derivation material.

In summary, three evasion strategies are blocked:

Evasion strategy	Blocked by	Mechanism
E1: Unitary-only protocol	R1	Paper 5 mandates Lüders
E2: Coherence-protected info	R1	Lüders destroys coherenc
E3: Equilibrium coherence	R2	Gibbs state has zero cohe

D. Three-link chain: self-modeling to entropy gradient

We now assemble the central result of the paper's thermodynamic argument: a three-link chain connecting self-modeling to the need for an entropy gradient. Each link is labeled by its epistemic status.

Proposition 3 (Chain theorem)**.** *Let $(B, M, \&)$ be a self-modeling composite on a finite SWAP lattice in thermal contact with an environment at temperature T. Then:*

Link 1 [THEOREM]. *Maintaining $I(B;M) > 0$ requires work $W_{\rm cycle} \geq k_{\rm B}T\,I(B;M)$ per cycle* (Theorem 2).

Link 2 [STANDARD PHYSICS]. *Performing work $W > 0$ per cycle requires the composite to be out of thermal equilibrium:*

$$F(\rho_{BM}) > F_{\rm eq} \quad \textit{when} \quad I(B;M) > 0\,, \tag{25}$$

where $F(\rho) = \langle H\rangle_\rho - T\,S_{\rm vN}(\rho)$ is the Helmholtz free energy and $F_{\rm eq}$ is its minimum at thermal equilibrium.

This is equivalent to a positive quantum relative entropy: the maximum extractable work is $W_{\max}^{\text{extract}} = k_{\text{B}}T\, D(\rho_{BM}\|\rho_{\text{eq}}) \geq 0$.

Link 3 [PHYSICAL ARGUMENT + A3]. *A finite system out of thermal equilibrium will equilibrate on a finite timescale unless it has access to a low-entropy reservoir. By Sec. II, the SWAP dynamics drives* $I(B;M) \to 0$ *geometrically* (Eq. (15)). *To sustain self-modeling beyond the exhaustion time, the system requires* $S(t) < S_{\max}$*: an entropy gradient.*

The chain has no logical gaps: Link 1 outputs $W > 0$, which is Link 2's input; Link 2 outputs $F > F_{\text{eq}}$, which is Link 3's input; Link 3 outputs the requirement for an entropy gradient.

Corollary 4 (Exhaustion bound)**.** *A self-modeler with initial free energy surplus* $\Delta F = F(\rho_0) - F_{\text{eq}}$ *can sustain at most*

$$N_{\text{cycles}} \lesssim \frac{\Delta F}{k_{\text{B}}T\, I(B;M)} \tag{26}$$

self-modeling cycles before reaching thermal equilibrium, at which point $I(B;M) = 0$ *and* $\rho = 0$.

a. Assumption A3 (strongest assumption). Link 3 relies on the physical assumption that a closed or semi-closed finite system equilibrates on finite timescales. This is standard statistical mechanics for systems with $t \ll t_{\text{rec}}$, but it is the strongest assumption in the chain. It can fail if the system is coupled to an infinite reservoir that never equilibrates (an idealization, not a physical scenario), if the universe has an infinite-dimensional Hilbert space (speculative), if the system exhibits many-body localization (MBL)—established in 1D disordered interacting chains but controversial in higher dimensions—or if it is integrable, thermalizing to a generalized Gibbs ensemble (GGE) rather than the thermal state. Neither MBL nor GGE is expected for the SWAP lattice (which is neither disordered nor integrable), but they represent concrete finite-system failure modes of A3 beyond the idealized scenarios listed above. A3 can also fail if there exist sources of free energy that do not ultimately derive from a low-entropy initial condition (unknown physics). Links 1 and 2 remain valid regardless of whether A3 holds.

b. Assumption A4 (model choice). The SWAP lattice dynamics $H = JF$ from Ref. [17] enters only through the quantitative equilibration rate. Replacing SWAP with any other thermalizing dynamics changes the timescale of convergence but not the qualitative conclusion: any finite system in thermal contact equilibrates.

E. Explicit non-claims

To prevent overclaiming, we state what the chain does *not* establish:

1. The chain does not derive the Past Hypothesis. It shows that self-modeling *requires* an entropy gradient, but does not explain why the initial entropy was low.
2. The chain does not derive the Second Law. Link 2 *uses* the Second Law as input (standard physics).
3. The chain does not predict the magnitude of the initial entropy of the universe.
4. The chain does not extend to infinite-dimensional systems without additional analysis. Assumption A1 (finite-dimensional QM) is required for the Reeb–Wolf quantum Landauer bound.
5. The connection from Link 3 to cosmology is a physical argument (assumption A3), not a mathematical theorem.

The status of the Past Hypothesis is elevated from "observed feature" to "necessary condition for self-modeling," but it remains an input to the argument, not an output.

With the thermodynamic cost of self-modeling established, we turn in Sec. IV to a second consequence of the self-modeling framework: the chirality from the exceptional Jordan algebra requires the emergent spacetime to carry a time-orientation, the same geometric structure that supports the thermodynamic arrow of time.

IV. CHIRALITY REQUIRES TIME-ORIENTATION

Paper 7 showed that a single choice of unit imaginary octonion $u \in S^6$ determines both the Standard Model gauge group and its chiral representation [6]. That derivation is purely algebraic: the Euclidean Clifford algebra Cl(6) decomposes the Spin(10) Weyl spinor into left- and right-handed sectors without reference to spacetime geometry. We now show that *physical realization* of this chirality—as spacetime Weyl spinors coupling to a chiral gauge theory—requires the emergent spacetime to carry a time-orientation. This is the new contribution of the present paper: the algebraic choice u not only builds particle physics structures (constructive) but also constrains spacetime geometry (constraint).

A. The chirality operator in Lorentzian signature

Consider an even-dimensional Lorentzian manifold of signature $(d-1, 1)$, with Clifford generators satisfying

$$\{\Gamma_\mu, \Gamma_\nu\} = 2\,\eta_{\mu\nu}\,, \tag{27}$$

where $\eta = \text{diag}(+1, -1, \ldots, -1)$ is the mostly-minus metric. In particular, $\Gamma_0^2 = +1$ (timelike) and $\Gamma_i^2 = -1$ for $i = 1, \ldots, d-1$ (spacelike).

The chirality operator (volume form) in $\mathrm{Cl}(d-1,1)$ is

$$\Gamma_* = i^k\,\Gamma_0\,\Gamma_1\cdots\Gamma_{d-1}\,, \tag{28}$$

where the integer k is chosen so that $(\Gamma_*)^2 = +1$. For $d = 4$ this gives the standard $\gamma_5 = i\,\Gamma_0\Gamma_1\Gamma_2\Gamma_3$ of four-dimensional physics.

The Weyl (chirality) projectors are

$$P_L = \frac{1+\Gamma_*}{2}\,, \qquad P_R = \frac{1-\Gamma_*}{2}\,. \tag{29}$$

These are orthogonal idempotents ($P_L+P_R = 1$, $P_LP_R = 0$), and they decompose the Dirac spinor into Weyl spinors: $S = S_L \oplus S_R$ with $S_L = P_LS$ and $S_R = P_RS$.

B. Time-reversal flips chirality

Time reversal acts on the Clifford generators as

$$T\colon\ \Gamma_0 \to -\Gamma_0\,, \qquad \Gamma_i \to \Gamma_i \quad (i = 1,\ldots,d{-}1)\,. \tag{30}$$

Proposition 5 (Chirality flip)**.** *For any even $d = 2m$, the chirality operator satisfies*

$$T(\Gamma_*) = -\Gamma_*\,. \tag{31}$$

Proof. The prefactor i^k is unaffected by T. The product $\Gamma_0\Gamma_1\cdots\Gamma_{d-1}$ contains Γ_0 exactly once, so under T it acquires precisely one factor of (-1):

$$\Gamma_0\Gamma_1\cdots\Gamma_{d-1} \xrightarrow{T} (-\Gamma_0)\Gamma_1\cdots\Gamma_{d-1} = -\Gamma_0\Gamma_1\cdots\Gamma_{d-1}\,.$$

Hence $T(\Gamma_*) = i^k\cdot(-\Gamma_0\Gamma_1\cdots\Gamma_{d-1}) = -\Gamma_*$. □

Corollary 6. *Under time reversal, the Weyl projectors exchange:*

$$P_L \xrightarrow{T} P_R\,, \qquad P_R \xrightarrow{T} P_L\,. \tag{32}$$

The proof is immediate from Eq. (31): replacing $\Gamma_* \to -\Gamma_*$ in (29) interchanges P_L and P_R.

We verify the chirality flip explicitly in three dimensions relevant to this program:

$d = 4$ (physical spacetime). $\gamma_5 = i\,\Gamma_0\Gamma_1\Gamma_2\Gamma_3$, $(\gamma_5)^2 = +1$ ✓. Under T: $\gamma_5 \to i(-\Gamma_0)\Gamma_1\Gamma_2\Gamma_3 = -\gamma_5$ ✓.

$d = 2$ (simplest non-trivial case). $\Gamma_* = \Gamma_0\Gamma_1$, $(\Gamma_*)^2 = +1$ ✓. Under T: $\Gamma_* \to -\Gamma_0\Gamma_1 = -\Gamma_*$ ✓.

$d = 10$ (the Spin(10) setting of the $h_3(\mathbb{O})$ program). $\Gamma_* = \Gamma_0\Gamma_1\cdots\Gamma_9$, $(\Gamma_*)^2 = +1$ ✓. Under T: $\Gamma_* \to -\Gamma_*$ ✓.

The result holds for all even-dimensional *Lorentzian* Clifford algebras $\mathrm{Cl}(d{-}1,1)$ because Γ_0 (the unique timelike generator) appears exactly once in the ordered product (28), regardless of d. In Euclidean signature there is no timelike direction and the proposition does not apply.

C. Internal vs. spacetime chirality

Paper 7's Clifford algebra $\mathrm{Cl}(6)$ and the spacetime Clifford algebra $\mathrm{Cl}(3,1)$ are distinct objects with distinct chirality operators:

Internal chirality (ω_6, Euclidean). The volume form of the Euclidean algebra $\mathrm{Cl}(6)$, with generators $\gamma_1,\ldots,\gamma_6$ satisfying $\{\gamma_i,\gamma_j\} = 2\delta_{ij}$, is [6]

$$\omega_6 = \gamma_1\gamma_2\gamma_3\gamma_4\gamma_5\gamma_6\,, \qquad \omega_6^2 = -1\,, \qquad (i\,\omega_6)^2 = +1\,. \tag{33}$$

The chirality operator $i\,\omega_6$ decomposes the 16-dimensional Weyl spinor S_{10}^+ into two 8-dimensional sectors carrying the Pati-Salam representations $(\mathbf{4},\mathbf{2},\mathbf{1})$ and $(\bar{\mathbf{4}},\mathbf{1},\mathbf{2})$. This decomposition is a *representation-theoretic* fact: it determines which SM quantum numbers each state carries. It is purely algebraic and involves no spacetime geometry.

Spacetime chirality (γ_5 or Γ_*, Lorentzian). The chirality operator of $\mathrm{Cl}(3,1)$ (or $\mathrm{Cl}(9,1)$) determines which Lorentz representation each fermion carries: left-handed spinors transform as $(\frac{1}{2},0)$ under $\mathrm{SL}(2,\mathbb{C})$, right-handed as $(0,\frac{1}{2})$. This operator involves Γ_0 and therefore *requires time-orientation* for a globally consistent definition (Proposition 5).

The correlation that defines the chiral SM. The Standard Model is a chiral gauge theory: its defining feature is that left-handed fermions couple differently from right-handed ones under $\mathrm{SU}(2)_L$. This means a fermion labeled "left-handed" must be *simultaneously*:

1. in the $(\mathbf{4},\mathbf{2},\mathbf{1})$ representation under Pati-Salam (from the internal ω_6), *and*
2. a left-handed Weyl spinor under the Lorentz group (from the spacetime γ_5).

The internal chirality alone assigns representation labels; the spacetime chirality determines propagation and coupling. The full chiral Standard Model requires both chiralities to be defined and correlated—and the spacetime part requires time-orientation.

D. Conditions for Weyl spinor bundles

We now state precisely which geometric structures a manifold needs to support Weyl spinor bundles. A globally consistent Γ_* with definite sign requires two independent data:

1. **Space-orientation**: to fix the sign from the ordered spatial product $\Gamma_1\cdots\Gamma_{d-1}$. A permutation of two generators flips the sign.
2. **Time-orientation**: to fix the sign from Γ_0. Reversing which direction is "future-pointing" sends $\Gamma_0 \to -\Gamma_0$ and hence $\Gamma_* \to -\Gamma_*$.

A Lorentzian spin manifold (M, g) admits Weyl spinor bundles $S(M) = S_L(M) \oplus S_R(M)$ if and only if M is both time-oriented and space-oriented [10]. Conversely, if both orientations are fixed, the ordered frame $\{e_0, e_1, \ldots, e_{d-1}\}$ (with e_0 future-pointing and $\{e_1, \ldots, e_{d-1}\}$ positively oriented) is globally defined up to proper orthochronous Lorentz transformations $\mathrm{SO}^+(d-1, 1)$, which preserve the sign of Γ_*.

We assume that the emergent spacetime of Paper 6 provides these structures:

Assumption A6 (continuum limit). *The continuum limit of Paper 6's self-modeling lattice yields a smooth Lorentzian manifold* (M, g).

Assumption A7 (Lorentzian signature). *The emergent spacetime has Lorentzian signature, as implied by Paper 6's causal structure from Lieb-Robinson bounds [17].*

Under these assumptions, the lattice structure provides all necessary geometric data. A d-dimensional regular lattice has d globally defined basis vectors, which in the continuum limit become a global frame (a trivialization of the tangent bundle). A global frame is *stronger* than each of the conditions it implies:

$$\text{framing} \implies \text{spin structure} \implies \text{orientability}\,, \tag{34}$$

where neither implication reverses.[1] Paper 6's lattice additionally provides time-orientation from Hamiltonian evolution (the sign of e^{-iHt} distinguishes $t > 0$ from $t < 0$ [17]) and space-orientation from the lattice ordering. Therefore all four conditions for Weyl spinor bundles—spin structure, space-orientation, time-orientation, and even-dimensionality—are satisfied.

E. Three consequences of a single choice

We are now in a position to extend Paper 7's "one choice, two consequences" theorem. Paper 7 showed that the single algebraic input $u \in S^6$ determines both the gauge group and the chiral representation [6]. The results of this section add a third consequence: the chirality from u imposes a geometric constraint on the emergent spacetime.

Theorem 7 (Three consequences of u)**.** *Let $h_3(\mathbb{O})$ be the exceptional Jordan algebra with observer idempotent E_{11}. Assume:*

(P1) $h_3(\mathbb{O})$ *is the universe algebra* (Gap A).

(P2) *The observer selects E_{11}, breaking $F_4 \to \mathrm{Spin}(9)$* (Gap B1).

(P3) *A complex structure $u \in S^6 \subset \mathrm{Im}(\mathbb{O})$ is chosen* (Gap B2).

(P4) *The complexification principle holds: $V_{1/2} \otimes_{\mathbb{R}} \mathbb{C} = S_{10}^+$* (Gap C).

(A6) *The continuum limit of Paper 6's lattice yields a smooth manifold.*

(A7) *The emergent spacetime has Lorentzian signature.*

Then the single choice of u simultaneously determines:

(a) **Gauge group** (constructive): *The F_4 intersection gives $\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, $\dim = 12$ [2, 6].*

(b) **Chirality** (constructive): *u defines $\mathrm{Cl}(6) \subset \mathrm{Cl}(10)$, whose volume form ω_6 selects the LEFT chiral embedding [3, 6]:*

$$\mathbf{16} \;\to\; (\mathbf{4}, \mathbf{2}, \mathbf{1}) \;\oplus\; (\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2})\,. \tag{35}$$

(c) **Time-orientation requirement** (constraint): *The chirality from* (b) *can only be physically realized as spacetime Weyl spinors on a time-oriented Lorentzian manifold. Therefore u creates a necessity for time-orientation on the emergent spacetime.*

Proof. Consequences (a) and (b) are proved in Paper 7 (Theorems 4.1 and 4.2 of [6]). For (c), the logical chain is:

$$u \xrightarrow{\text{alg}} W = u^\perp \xrightarrow{\text{alg}} \mathrm{Cl}(6) \xrightarrow{\text{alg}} \omega_6 \xrightarrow{\text{rep}} \text{L/R decomp} \xrightarrow{\text{phys}} \tag{36}$$

Steps 1–4 are pure algebra (from u to the L/R decomposition via ω_6). Step 5 is the physical identification of the algebraic L/R labels with spacetime Weyl spinor types. Step 6 is Proposition 5: the spacetime chirality operator Γ_* involves Γ_0, so a globally consistent Weyl decomposition requires time-orientation. The argument is not circular: steps 1–4 assume no geometry, and time-orientation appears only as a *conclusion*, not a premise. □

Remark 8 (Nature of consequence (c)). Consequences (a) and (b) are *constructive*: u builds the gauge group and the chiral representation through explicit algebraic procedures. Consequence (c) is a *constraint*: u imposes a geometric requirement on the emergent spacetime but does not itself select a specific time-orientation. The particular "arrow of time"—which direction is the future—comes not from u but from the thermodynamic gradient developed in Section V.

Remark 9 (Assumptions A6–A7 are new). Assumptions P1–P4 are inherited from Paper 7; they suffice for consequences (a) and (b), which are purely algebraic. Assumptions A6–A7 are needed only for consequence (c) and its satisfaction: they place the algebraic chirality into a geometric context. If A6 or A7 fail, consequences (a) and (b) are unaffected, and the three-consequence theorem reduces to Paper 7's two-consequence theorem.

[1] Counterexamples showing strictness: S^4 is spin but not frameable ($\chi(S^4) = 2 \neq 0$); $\mathbb{C}P^2$ is oriented but not spin ($w_2 \neq 0$).

That Paper 6's emergent spacetime satisfies the constraint is the consistency check: under assumptions A6–A7, the lattice framing provides a spin structure, the lattice ordering provides space-orientation, and Hamiltonian evolution provides time-orientation [17]. The chirality from (b) can therefore be physically realized.

The key structural insight is that the same choice u that determines particle physics content (gauge group and chirality) also constrains spacetime geometry (time-orientation). This cross-paper link—particle physics from Paper 7 constraining spacetime from Paper 6—was implicit in the program but has not been stated before. It will become central in the next section, where we show that the time-orientation required by chirality is the *same* geometric prerequisite needed for the thermodynamic arrow of time.

V. ENTROPY GRADIENT THEOREM AND GAP C RESOLUTION

We now bring together the entropy dynamics of Sec. II, the Landauer bound of Sec. III, and the chirality–time-orientation link of Sec. IV to establish the central results of this paper. The argument has four parts. First, we prove that every self-modeler requires an entropy gradient $S(t) < S_{\max}$—what we call the *entropy gradient theorem*—via three independent routes (Sec. V A). Second, we identify two independent chains that converge on time-orientation as a shared geometric prerequisite (Sec. V B). Third, we state the narrowed Gap C result as a formal theorem with epistemic labels (Sec. V C). Fourth, we present the updated master theorem and nine-link chain with Link 6 annotated (Secs. V D–V E).

A. Entropy gradient theorem

Theorem 10 (Entropy gradient)**.** *Let $(B, M, \&)$ be a self-modeling composite on a finite SWAP lattice in thermal contact with an environment at temperature T, with $\dim \mathcal{H}_B = d_B$, $\dim \mathcal{H}_M = d_M$, and faithful tracking $I(B;M) > 0$. Then $S(t) < S_{\max} \equiv \ln(d_B\, d_M)$: the composite must be out of thermal equilibrium, with entropy strictly below its maximum.*

We prove this via three independent routes, each using a different subset of assumptions. Their convergence on the same conclusion strengthens the result.

a. Route A (Direct chain). This route assembles the chain theorem of Sec. III D into a single implication.

1. **Self-modeling ⇒ work.** Theorem 2 gives $W_{\min} = k_{\mathrm{B}}T\, I(B;M) > 0$ per update cycle. [THEOREM, Sec. III B]

2. **Work ⇒ non-equilibrium.** Performing work $W > 0$ requires free energy above the equilibrium value: $F(\rho_{BM}) > F_{\mathrm{eq}}$. Equivalently, the extractable work $k_{\mathrm{B}}T\, D(\rho_{BM}\|\rho_{\mathrm{eq}}) > 0$ implies $\rho_{BM} \neq \rho_{\mathrm{eq}}$. [STANDARD PHYSICS]

3. **Non-equilibrium ⇒ entropy gradient.** A finite system out of equilibrium will equilibrate on a timescale set by the SWAP dynamics (Eq. (15)). To sustain self-modeling beyond the exhaustion bound (26), the system must access a low-entropy reservoir: $S(t) < S_{\max}$. [PHYSICAL ARGUMENT + A3]

The chain is:

$$I(B;M) > 0 \xrightarrow{\text{Thm. 2}} W > 0 \xrightarrow{\text{2nd Law}} F > F_{\mathrm{eq}} \xrightarrow{\text{A3}} S(t) < S_{\max} \tag{37}$$

Each link's output is the next link's input, and no link uses the conclusion of a later link.

b. Route B (Boundary argument). This route avoids the Landauer bound entirely, using only finitude and monotonicity.

1. **Entropy is bounded.** Finite-dimensional quantum mechanics gives $0 \leq S(\rho_{BM}) \leq \ln(d_B\, d_M) = S_{\max}$. [A1]

2. **Entropy is non-decreasing.** Under repeated interaction with fresh maximally mixed bath qubits, the monotonicity chain (14) holds. On a macroscopic lattice ($N \gg 1$), effective monotonicity holds for $t \ll t_{\mathrm{rec}} \sim e^{c\cdot 2^N}$ (Proposition 1). [THEOREM, Sec. II B]

3. **Self-modeling requires $S < S_{\max}$.** At $S = S_{\max}$ the state is maximally mixed, so $I(B;M) = 0$ (Eq. (20)) and $\rho = 0$. Therefore $\rho > 0$ implies $S(t) < S_{\max}$. [DEFINITION, Eq. (18)]

Combining: S is bounded, non-decreasing, and currently below its maximum. Therefore for all $t_0 < t$,

$$S(t_0) \;\leq\; S(t) \;<\; S_{\max}\,. \tag{38}$$

The block possesses a low-entropy past.

c. Route C (ρ-selection). This route requires the fewest assumptions. The contrapositive of three established results gives:

$$\rho > 0 \implies I(B;M) > 0 \implies \rho_{BM} \neq \rho_{\mathrm{eq}} \implies S(t) < S_{\max}\,. \tag{39}$$

Step 1 is the definition of ρ (Eq. (18): $\rho = 0$ when $I(B;M) = 0$). Step 2 is the equilibrium result $I(B;M)^{\mathrm{eq}} = 0$ (Eq. (20)). Step 3 is $S(\rho_{\mathrm{eq}}) = S_{\max}$. The ρ-weighted ensemble is therefore concentrated on blocks with $S < S_{\max}$: the experiential measure sees only non-equilibrium blocks.

Comparison of routes.—Table I compares the three routes. Route A provides the strongest quantitative content (the Landauer cost per cycle). Route B provides the simplest logic (bounded monotone sequences). Route C requires the fewest assumptions (only the definition of ρ and the equilibrium result). Their convergence from different starting points makes the entropy gradient theorem robust against the failure of any single assumption.

TABLE I. Three routes to the entropy gradient theorem. Each route proves $S(t) < S_{\max}$ using a different subset of assumptions. "An" labels refer to the assumption register in Table III.

	Route A	Route B	Route C
Key input	Landauer bound	Finitude + monotonicity	$\rho > 0 \Leftrightarrow I > 0$
Assumptions	A1, A2, A3, A4	A1, A3	A1, A5
Epistemic chain	Thm. + std. phys. + A3	Thm. (Lindblad) + arithmetic Def. + thm. + contrapositive	
Gives $W_{\min}$?	Yes	No	No
Uses Landauer?	Yes	No	No

B. Two chains and their convergence

The entropy gradient theorem and the chirality–time-orientation link of Sec. IV provide two *independent* chains of reasoning. We present each, then show that they converge on a common geometric prerequisite: time-orientation.

a. Chain 1 (all self-modelers). The entropy gradient theorem (Theorem 10) establishes that *every* self-modeling composite requires an entropy gradient, regardless of chirality or complexification. The logic, proved via three independent routes in Sec. V A, is:

$$I(B;M) > 0 \xrightarrow{\text{Landauer}} W \geq k_{\mathrm{B}} T\, I(B;M) \xrightarrow{\text{2nd Law}} \text{non-equilibrium} \xrightarrow{\text{Lem. 9}} S(t) < S_{\max}. \tag{40}$$

This chain holds for all self-modelers on any block of structure space. It makes no reference to chirality, complexification, or the Standard Model. It uses only assumptions A1–A3 (Table III).

The entropy gradient defines a temporal direction: the "future" is the direction of entropy increase, and the "past" is the direction of lower entropy. On an emergent Lorentzian manifold (A6–A7), this thermodynamic arrow requires a time-orientation—a continuous choice of future-pointing timelike vector field—to distinguish the two null half-cones at every point. This last step uses A6–A7 and is a physical identification (thermodynamic arrow → geometric time-orientation), not a mathematical theorem; it is the standard assumption that the emergent spacetime's causal structure is compatible with its thermodynamics.

b. Chain 2 (SM-like observers). Standard Model physics is defined by chiral fermions. The chain from complexification to chirality to time-orientation is:

$$\text{SM-like observer} \xrightarrow{\text{def.}} \text{chirality} \xrightarrow{[6]} \text{complexification} \xrightarrow{\text{CHIR-01}} \text{time-orientation}. \tag{41}$$

Each arrow is a necessary condition, reading right-to-left:

- **Chirality requires complexification.** On real $V_{1/2} = \mathbb{O}^2$, the Cl(6) volume form ω_6 has no eigenstates giving the left–right decomposition: the eigenvalue equation $i\,\omega_6\,|\psi\rangle = \pm|\psi\rangle$ requires complex coefficients. Without complexification, chirality is not defined. [ALGEBRA; [6]]

- **Chirality requires time-orientation.** The three-consequence theorem (Theorem 7, consequence (c)) establishes that the physical Weyl decomposition $S = S_L \oplus S_R$ requires both internal chirality and a time-orientation for a globally consistent sign of the spacetime chirality operator Γ_* (Proposition 5, Corollary 6). [GEOMETRY; Sec. IV]

This chain uses assumptions A6–A7 in addition to the algebraic results of [6].

c. Logical status of the two chains. The direction of implication is critical. The three-consequence theorem proves *chirality ⇒ time-orientation is required*. Its valid contrapositive is *no time-orientation ⇒ no chirality*. Crucially, the *converse*—no chirality ⇒ no time-orientation—does *not* follow. A non-complexified block can lack chirality while still possessing time-orientation from other sources.

This is why the two chains are independent. Chain 1 establishes entropy gradients (and hence a temporal direction) for all self-modelers without invoking chirality. Chain 2 shows that SM-like observers additionally require chirality, which in turn requires both complexification and time-orientation. Neither chain implies the other.

d. Structural convergence. The two chains converge on time-orientation as a shared geometric prerequisite:

- **Chain 1:** the entropy gradient defines a temporal direction (low-entropy past → high-entropy future), which on a Lorentzian manifold corresponds to a time-orientation.

- **Chain 2:** chirality requires time-orientation for the Weyl decomposition (Theorem 7(c)).

The single algebraic choice $u \in S^6$ does double duty: it determines the gauge group and chirality (Paper 7 [6]) *and* creates a requirement for the same time-orientation that the thermodynamic arrow provides. This convergence is a structural consistency check: the framework's particle physics content and its thermodynamic content both require a geometric property (time-orientation) that the emergent spacetime automatically provides.

It is *not*, however, a proof that non-complexified blocks have $\rho = 0$. A block without chirality can still possess time-orientation (from Chain 1 alone), sustain an entropy gradient, and support non-SM self-modelers. The convergence constrains SM-like observers; it does not exclude all observers from non-complexified blocks.

Schematically, the two-chain structure is:

Chain 1: self-modeling ⇒ $S < S_{\max}$ ⇒ time-orientation (all se

Chain 2: SM-like ⇒ chirality ⇒ complexification + time-orienta

(42)

C. Gap C: narrowed

We are now in a position to state the central result regarding Gap C. The result is a *narrowing*, not a full resolution: complexification is necessary for SM-like observers, but we cannot exclude non-SM self-modelers in non-complexified blocks.

Theorem 11 (SM-observer complexification requirement)**.** *Let $\mathcal{S}$ be structure space with experiential measure μ_ρ. Let $\mathcal{B}$ be a block of $\mathcal{S}$ supporting a C^*-observer ($M_n(\mathbb{C})^{\mathrm{sa}}$) probing $h_3(\mathbb{O})$. Suppose assumptions A1–A7 hold* (Table III)*. If $V_{1/2}(\mathcal{B})$ is not complexified* (*remains real $\mathbb{O}^2$*)*, then:*

(a) *$\mathcal{B}$ has* ***no chiral spinors****: the* Cl(6) *volume form ω_6 has no eigenstates in real $V_{1/2}$ that give the L/R decomposition.* [ALGEBRA; [6]]

(b) *$\mathcal{B}$ has* ***no SM gauge group****: without complexification, the breaking chain* $\mathrm{Spin}(10) \to \mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R \to \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ *is not realized.* [ALGEBRA; [6]]

(c) *Chirality* ***requires time-orientation****: the physical Weyl decomposition $S = S_L \oplus S_R$ requires a time-orientation for a globally consistent sign of Γ_*.* [GEOMETRY; Sec. IV]

(d) *Self-modelers* ***independently require entropy gradients****: $I(B;M) > 0$ implies $S(t) < S_{\max}$ via Theorem 10, with no reference to chirality.* [THERMODYNAMICS; Sec. V A]

The convergence of (c) *and* (d) *on time-orientation means that within $h_3(\mathbb{O})$, the particle physics content* (*gauge group, chirality*) *and the thermodynamic content* (*entropy gradient, arrow of time*) *share a common geometric prerequisite.*

Consequently, every block with SM-like observers (*chiral fermions, SM gauge group*) *has complexified $V_{1/2}^{\mathbb{C}} = S_{10}^+$.*

Proof. Part (a) is pure representation theory: on the real space $V_{1/2} = \mathbb{O}^2$, the operator $i\,\omega_6$ is not defined (i requires complex scalars), so the eigenvalue equation $i\,\omega_6|\psi\rangle = \pm|\psi\rangle$ has no solutions. The left–right decomposition (35) therefore does not exist. This is proved in Ref. [6].

Part (b) follows from (a): the gauge symmetry breaking chain $\mathrm{Spin}(10) \to \mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R \to \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ requires the complexified representation S_{10}^+ for its chiral content. Without complexification, the Spin(10) structure acting on real $V_{1/2}$ does not decompose into SM representations. This is established in Ref. [6].

Part (c) follows from Theorem 7(c): the physical realization of chirality as spacetime Weyl spinors requires both internal chirality [from (a)] and time-orientation (Proposition 5). The direction of implication is: chirality $\Rightarrow$ time-orientation required. The valid contrapositive is: no time-orientation $\Rightarrow$ no chirality.

Part (d) is the entropy gradient theorem (Theorem 10), proved via three independent routes in Sec. V A. This part is logically independent of (a)–(c): it applies to all self-modelers regardless of chirality or complexification.

The convergence statement follows from (c) and (d): both the chirality requirement and the self-modeling requirement point to time-orientation, but through independent logical paths.

The final statement (SM-like $\Rightarrow$ complexified) follows from (a) by contrapositive: no complexification $\Rightarrow$ no chirality $\Rightarrow$ not SM-like. □

Remark 12 (Selection, not algebraic forcing). The complexification of $V_{1/2}$ for SM-like observers is a *selection effect*, not an algebraic derivation. Phase 22 of this program established that four independent algebraic routes to complexification all fail: the $V_1 = \mathbb{R} \cdot E_{11}$ bottleneck in the Peirce decomposition is a genuine obstruction [5]. Theorem 11 does not override this negative result. Non-complexified blocks exist in structure space; they are not algebraically forbidden. SM-like observers require complexification, but the theorem does not establish $\rho = 0$ for all non-complexified blocks: a non-complexified block could in principle support non-SM self-modelers with $\rho > 0$.

a. What is proved and what is not. It is important to state the epistemic status precisely:

- **Proved:** Complexification is necessary for SM-like observers (chiral fermions, SM gauge group).
- **Proved:** All self-modelers require entropy gradients, independent of chirality.
- **Proved:** Chirality and thermodynamics share time-orientation as a common geometric prerequisite.
- **Not proved:** $\rho = 0$ for all non-complexified blocks. A non-complexified block could have time-orientation, an entropy gradient, and non-SM self-modelers with $\rho > 0$.

The last point does not undermine the program: the program explains *our* physics (SM particle content, chiral fermions, Pati–Salam unification), and for that physics, complexification is necessary. Non-SM blocks, if they exist, are inaccessible under L4—different structure-space points do not interact.

b. Gap C: narrowed. Paper 7's nine-link chain [6] identifies Gap C as the complexification step $V_{1/2} \to V_{1/2}^{\mathbb{C}} = S_{10}^+$ (Link L6). Theorem 11 *narrows* this gap: among all blocks supporting SM-like observers, $V_{1/2}$ is complexified. The gap is not fully resolved because we cannot exclude non-SM self-modelers in non-complexified blocks. The narrowing is via the experiential measure restricted to the SM sector, not via algebraic necessity. This is the appropriate level of progress given Phase 22's proof that algebraic closure is impossible [5].

c. Status of the Past Hypothesis. The entropy gradient theorem (Theorem 10) shows that self-modeling *requires* $S(t) < S_{\max}$. Combined with the entropy monotonicity of Sec. II, this implies a low-entropy boundary condition at earlier times (Eq. (38)). The Past Hypothesis—the cosmological observation that the universe began in a state of extraordinarily low entropy [11, 12]—is thereby *elevated* from "observed feature of the universe" to "necessary condition for self-modeling." It is not *derived*: the entropy gradient theorem is silent on *why* the initial entropy was low. It establishes only that it *must* have been low for blocks with $\rho > 0$ to exist.

D. The v7.0 master theorem

We collect the results of this paper and its predecessors into a single statement.

Theorem 13 (Arrow of time and SM-observer complexification)**.** *Under assumptions A1–A7* (Table III)*, the following hold for self-modeling composites* $(B, M, \&)$ *on a finite SWAP lattice probing* $h_3(\mathbb{O})$*:*

(I) **Entropy gradient.** *Self-modelers require an entropy gradient:* $S(t) < S_{\max}$*. This is established via three independent routes* (Table I)*, each using a different assumption subset. This holds for all self-modelers regardless of chirality or complexification.*

(II) **SM-observer complexification.** *SM-like observers within* $h_3(\mathbb{O})$ *require complexified* $V_{1/2}$*. Non-complexified blocks have no chiral spinors and no SM gauge group* (*Theorem 11(a)–(b)*)*.*

(III) **Gap C status: narrowed.** *Complexification is necessary for SM-like observers* (*selection effect, not algebraic necessity*)*. The gap remains open for non-SM self-modelers: a non-complexified block could in principle support observers with* $\rho > 0$ *but without chiral fermions or the SM gauge group.*

Corollary 14 (Chirality–thermodynamics nexus)**.** *The chirality of the Standard Model and the thermodynamic arrow of time share a common geometric prerequisite:* ***time-orientation*** *of the emergent spacetime. Chirality requires time-orientation for the physical Weyl decomposition* (Theorem 7(c))*. The arrow of time requires time-orientation to define the direction of entropy increase* (Theorem 10)*. Both are consequences of the self-modeling framework: the choice* $u \in S^6$ *that determines particle physics* (*gauge group and chirality [6]*) *simultaneously constrains spacetime geometry* (*time-orientation*)*, and this time-orientation is the same structure that supports the entropy gradient sustaining self-modeling. This convergence is structural, not causal: neither chain implies the other.*

E. Updated nine-link chain

Paper 7's nine-link chain from self-modeling to the chiral Standard Model [6] is updated with the results of this paper. Table II shows each link, its status, and the annotation from this work.

Of the nine links, six are proved algebraically (L1–L3, L5, L8, L9), one is assumed (L4, Gap B1), one is updated with a new geometric consequence (L7), and one is narrowed via selection (L6, this paper): necessary for SM-like observers, open for non-SM self-modelers.

F. Assumption register

Table III lists every assumption entering the master theorem (Theorem 13), ordered by severity. Assumption A3 is the strongest: it connects local thermodynamics to cosmological initial conditions and is the only assumption that, if it fails, weakens the "low-entropy past" conclusion of Route A. Assumptions A6–A7 are needed only for the SM-observer complexification requirement [part (II) of the master theorem]; the entropy gradient theorem [part (I)] survives their failure.

G. Explicit non-claims

To prevent overclaiming, we state what the results of this section do *not* establish.

1. **Gap C is not closed algebraically.** The algebraic obstruction ($V_1 = \mathbb{R} \cdot E_{11}$) found in Phase 22 [5] remains genuine. Theorem 11 narrows Gap C via the experiential measure restricted to SM-like observers, not by overriding the algebraic result.

2. $\rho = 0$ **is not proved for all non-complexified blocks.** A non-complexified block could possess time-orientation, sustain an entropy gradient, and support non-SM self-modelers with $\rho > 0$. Complexification is necessary for SM-like observers, not for all observers.

3. **No chirality does not imply no time-orientation.** The three-consequence theorem proves chirality $\Rightarrow$ time-orientation. The valid contrapositive is no time-orientation $\Rightarrow$ no chirality. The converse—no chirality $\Rightarrow$ no time-orientation—is logically invalid and is not used in any argument of this paper.

4. **This does not derive the Past Hypothesis.** The entropy gradient theorem shows that $S(t) < S_{\max}$ is *necessary* for self-modeling. It does not explain *why* the universe started with low entropy.

TABLE II. Updated nine-link chain. Links are numbered as in Paper 7 [6]. "Status" indicates the epistemic classification; "Annotation" records changes from this paper and its predecessors (Papers 5–7). Link L6 is narrowed via selection (Theorem 11); Link L7 is updated with the three-consequence theorem (Theorem 7).

Link	Statement	Status	Annotation
L1	Self-modeling composite $(B, M, \&)$ on $h_3(\mathbb{O})$	**Proved** [7]	—
L2	Observer is C^*-algebra $(M_n(\mathbb{C})^{\mathrm{sa}})$	**Proved** [7]	—
L3	Faithful tracking requires local tomography	**Proved** [7]	—
L4	$h_3(\mathbb{O})$ probed via rank-1 idempotent $e = E_{11}$	**Assumed** (Gap B1)	—
L5	Peirce decomposition: $V_1 \oplus V_{1/2} \oplus V_0$	**Proved** [6]	—
L6	$V_{1/2}$ complexifies to S^+_{10}	**Narrowed via selection**	Thm. 11: necessary for SM-like; ope
L7	$u \in S^6$ determines gauge + chirality + time-orient.	**Proved + updated**	Thm. 7: third consequence (c)
L8	$\mathrm{Spin}(10) \to \mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R \to \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$	**Proved** [6]	—
L9	Chiral SM fermions from Cl(6) action	**Proved** [6]	—

TABLE III. Assumption register for the master theorem (Theorem 13). Assumptions are ordered by decreasing severity (A3 strongest, A1/A5 weakest). "Used in" refers to the parts of the master theorem.

ID	Assumption	Source	Status	Used in
A1	Finite-dimensional QM	Paper 5 [7]	Axiom	(I) all routes; (II) all steps
A2	Thermal contact at temperature T	Standard [14]	Standard physics	(I) Route A, Link 1
A3	Finite system equilibrates on finite timescales	Stat. mech.	Physical argument	(I) Routes A,B; Link 3
A4	SWAP lattice dynamics $H = JF$	Paper 6 [17]	Model choice	(I) timescales only
A5	$\rho = I(B;M)(1 - I(B;M)/S_{\mathrm{vN}}(\rho_B))$	Paper 5 [7]	Axiom	(I) Route C; (II) step (d)
A6	Continuum limit yields smooth manifold	Paper 6 [17]	Assumption	(II) steps (a)–(c)
A7	Lorentzian signature $(d{-}1, 1)$	Paper 6 [17]	Physical	(II) steps (a)–(c)

5. **The Past Hypothesis is elevated, not derived.** Its status changes from "observed cosmological fact" to "necessary condition for self-modeling"—an observer-selection argument, not a first-principles derivation [11, 12].

6. **"Narrowed via selection" is weaker than "resolved via selection," which is weaker than "proved algebraically."** An algebraic proof of complexification would close Gap C unconditionally. A full selection resolution would show $\rho = 0$ for all non-complexified blocks. The narrowing establishes necessity for SM-like observers only.

With Gap C narrowed and the master theorem stated, we turn in Sec. VI to the testable consequences of the framework.

VI. QUANTITATIVE PREDICTIONS

The entropy gradient theorem (Sec. V) and the Landauer bound (Sec. III) together constrain the thermodynamic environment in which self-modeling composites can exist. We now extract quantitative predictions from these results. The predictions fall into two classes: *framework predictions* (P1–P6), which follow from assumptions A1–A7 alone and carry no model-dependent parameters, and *order-of-magnitude estimates* (P7–P9), which additionally require cosmological inputs (A8–A10). We compare the Landauer-derived entropy deficit with Penrose's gravitational entropy estimate [11], finding a gap of 94 orders of magnitude that we argue is a feature of the framework rather than a failure.

A. Minimum entropy deficit for self-modeling

From the exhaustion bound (Corollary 4), a self-modeler with initial free energy surplus $\Delta F = F(\rho_0) - F_{\mathrm{eq}}$ can sustain at most $N_{\mathrm{cycles}} \lesssim \Delta F/(k_{\mathrm{B}}T\, I(B;M))$ cycles. Since the available free energy is $\Delta F \approx k_{\mathrm{B}}T\,\Delta S$ with $\Delta S = S_{\max} - S$ the entropy deficit (in nats)—an approximation valid when the energy spectrum is much smaller than $k_{\mathrm{B}}T$ (the exact expression is $\Delta F = k_{\mathrm{B}}T\, D(\rho\|\rho_{\mathrm{eq}})$, where D denotes relative entropy)—the temperature cancels:

$$N_{\mathrm{exhaust}} = \frac{\Delta S}{I(B;M)}\,. \tag{43}$$

This result is purely informational: neither the bath temperature T nor the coupling constant J appears. For self-modeling to occur at all ($N_{\mathrm{exhaust}} \geq 1$), the initial entropy must satisfy

$$\boxed{S_{\mathrm{initial}} \leq S_{\max} - N\, I(B;M)} \tag{44}$$

for N update cycles. The minimum entropy deficit is therefore

$$\Delta S_{\min} = N\, I(B;M)\,. \tag{45}$$

This is an exact consequence of the Landauer bound (Theorem 2) and the definition of available work. It requires only assumptions A1 (finite-dimensional QM) and A2 (thermal contact).

B. Comparison with the Penrose entropy deficit

To evaluate Eq. (45) numerically, we introduce three cosmological inputs that are *external* to the framework.

a. A8 (cosmological entropy budget). The Bekenstein–Bousso bound on the cosmological horizon gives

$$S_{\max} \sim \frac{\pi R_H^2}{\ell_P^2} \sim 10^{122}\ \text{nats}\,, \tag{46}$$

where $R_H \sim 4.4 \times 10^{26}\,\text{m}$ is the Hubble radius and $\ell_P \sim 1.6 \times 10^{-35}\,\text{m}$ is the Planck length.

b. Penrose initial entropy. Penrose estimated the initial gravitational entropy of the observable universe from the Weyl curvature hypothesis [11]:

$$S_{\text{initial}} \sim 10^{88}\ \text{nats}\,, \tag{47}$$

yielding a Penrose entropy deficit $\Delta S_{\text{Penrose}} = S_{\max} - S_{\text{initial}} \approx 10^{122}$ nats (since $10^{122} \gg 10^{88}$).

c. A9 (neural self-modeling dimension) and A10 (update rate). A human-scale self-modeler with $\sim 10^{10}$ neural degrees of freedom at $\sim \ln 2$ nats each has mutual information $I(B;M) \sim 7 \times 10^9$ nats. At an update rate $\nu \sim 10$ Hz over a cosmic time $t_U \sim 4 \times 10^{17}$ s, the total number of cycles is $N \sim \nu\, t_U \sim 4 \times 10^{18}$.

The Landauer entropy deficit is then

$$\Delta S_{\text{Landauer}} = N\, I(B;M) \sim 4{\times}10^{18}{\times}7{\times}10^9 \sim 3{\times}10^{28}\ \text{nats}\,. \tag{48}$$

The ratio to Penrose's deficit is

$$\frac{\Delta S_{\text{Landauer}}}{\Delta S_{\text{Penrose}}} \sim \frac{3 \times 10^{28}}{10^{122}} \sim 10^{-94}\,. \tag{49}$$

The Landauer bound on self-modeling requires an entropy deficit of order 10^{28} nats. The observed deficit (as estimated by Penrose) is $\sim 10^{122}$ nats—larger by **94 orders of magnitude**.

This gap is a *feature*, not a failure, of the framework. The Landauer bound constrains the *information-processing* cost of self-modeling: how much entropy headroom a self-modeler must consume to maintain body–model correlations. Penrose's estimate constrains the *gravitational* entropy budget: how smooth the initial geometry must be relative to the black-hole-dominated equilibrium. These are different physical mechanisms. The enormous gap reflects the fact that gravitational entropy dominates the universe's entropy budget, while the thermodynamic cost of self-modeling is negligible by comparison.

The gap is robust to order-of-magnitude variations in the model-dependent parameters. Even if $I(B;M)$ is uncertain by a factor of 10^3 and N by a factor of 10^3, the Landauer deficit shifts to $\sim 10^{34}$ nats, leaving a gap of ~ 88 orders of magnitude.

C. Exhaustion timescale

The exhaustion timescale—the time before the entropy budget is consumed and self-modeling becomes thermodynamically impossible—is

$$\tau_{\text{exhaust}} = N_{\text{exhaust}}\, t_{\text{cycle}} = \frac{\Delta S}{I(B;M)}\, t_{\text{cycle}}\,. \tag{50}$$

Using the Penrose deficit $\Delta S \sim 10^{122}$ nats, $I(B;M) \sim 7 \times 10^9$ nats, and $t_{\text{cycle}} = 1/\nu \sim 0.1$ s:

$$\tau_{\text{exhaust}} \sim \frac{10^{122}}{7 \times 10^9} \times 0.1\ \text{s} \sim 10^{111}\ \text{s}\,. \tag{51}$$

Compared with the current age of the universe,

$$\frac{\tau_{\text{exhaust}}}{t_U} \sim \frac{10^{111}}{4 \times 10^{17}} \sim 10^{93}\,. \tag{52}$$

Self-modeling is thermodynamically permitted for a time $\sim 10^{93}$ times the present age of the universe. The entropy budget is far from exhausted; the universe is in the earliest stages of its thermodynamic lifetime as measured by the self-modeling capacity.

Note that τ_{exhaust} depends on the entropy budget (A8–A9) and the self-modeling parameters (A10), but does *not* depend on the bath temperature T, which cancels between the available work $k_{\text{B}} T\, \Delta S$ and the cost per cycle $k_{\text{B}} T\, I(B;M)$.

D. Experiential density profile

The experiential density [7],

$$\rho(I(B;M)) = I(B;M)\left(1 - \frac{I(B;M)}{S_B}\right), \tag{53}$$

where $S_B = S_{\text{vN}}(\rho_B)$ is the body entropy, is a concave parabola in $I(B;M)$ on the interval $[0,\, S_B]$. Setting $\mathrm{d}\rho/\mathrm{d}I = 1 - 2I/S_B = 0$ yields

$$I^* = \frac{S_B}{2}\,, \qquad \rho_{\max} = \frac{S_B}{4}\,. \tag{54}$$

The profile has three notable limits:

1. $I(B;M) = 0$ (no tracking): $\rho = 0$. At thermal equilibrium, the model contains no information about the body (Eq. (20)).

2. $I(B;M) = S_B$ (saturated tracking): $\rho = 0$. The pure-state constraint $S(\rho_{BM}) = 0$ enforces $I(B;M) = 2\,S_B$ only for pure joint states; for $I(B;M) = S_B$ the parabola returns to zero.

3. $N \to \infty$ (equilibration): The monotonicity chain (14) drives $I(B;M) \to 0$, so $\rho \to 0$ regardless of initial conditions.

Under the passive equilibration dynamics of Sec. II C, the mutual information decays geometrically: $I(N) \sim I_0\,(1-p)^N$, where $p = \sin^2(Jt)$ is the depolarizing parameter. The time dependence of ρ then takes the form

$$\rho(N) = I_0\,(1-p)^N \left(1 - \frac{I_0\,(1-p)^N}{S_B}\right). \tag{55}$$

If $I_0 > S_B/2$, this rises to the peak $\rho_{\max} = S_B/4$ (when I has decayed to $S_B/2$) and then falls to zero. If $I_0 \le S_B/2$, ρ decreases monotonically from the start. Both scenarios terminate at $\rho = 0$ as the system equilibrates.

E. Model-dependence register

The quantitative predictions P7–P9 rely on three assumptions beyond the framework (A1–A7). We record each, its role, and the sensitivity of predictions to its value.

ID	Parameter	Enters via	Sensitivity
A8	$S_{\max} \sim 10^{122}$ nats	Entropy budget ΔS	$\tau_{\text{exhaust}} \propto S_{\max}$
A9	$S_{\text{init}} \sim 10^{88}$ nats	Penrose comparison	Gap shifts by 2 orders
A10	$d_M \sim 10^{11}$, $\nu \sim 10$ Hz	$I(B;M)$, N in Eq. (48)	$\Delta S_{\text{L}} \propto I \cdot N$

These are *separate* from the framework assumptions A1–A7. Varying A8–A10 by several orders of magnitude in either direction shifts the Landauer deficit $\Delta S_{\text{Landauer}}$ accordingly but does not close the 94-order gap with Penrose. The framework predictions P1–P6 (Table IV) are independent of A8–A10.

F. Prediction summary

Table IV collects the predictions of the entropy gradient theorem and its consequences. Each entry is classified by strength and the assumptions it requires.

G. Non-claims

To guard against overclaiming, we state explicitly what the quantitative results do *not* establish.

NC-1.: The Landauer bound is not a prediction of the initial entropy of the universe. It provides a *necessary condition* ($\Delta S \gtrsim 10^{28}$ nats), not a determination. The observed deficit ($\sim 10^{122}$ nats) is vastly larger, and its origin lies in gravitational physics [11], not in the self-modeling framework.

NC-2.: The comparison with Penrose is not a derivation of $S_{\text{initial}} \sim 10^{88}$. The two constraints address different physics: Landauer constrains information-processing cost; Penrose constrains gravitational entropy via the Weyl curvature hypothesis. The comparison demonstrates internal consistency (the framework's bound is weaker), not explanatory power.

NC-3.: The exhaustion timescale $\tau_{\text{exhaust}} \sim 10^{111}$ s is not a prediction of the age of the universe. It quantifies how long the entropy budget permits self-modeling to continue, not how old the universe is.

NC-4.: The experiential density profile $\rho(I)$ is not computed to observational precision. The qualitative shape (peak at $I = S_B/2$, decay to zero at equilibrium) is exact given assumption A5; the mapping from update steps N to cosmological time is model-dependent (requiring A8–A10 and the specific equilibration dynamics from Sec. II).

The quantitative results of this section are best understood as consistency checks: the framework's thermodynamic constraints are compatible with—and vastly weaker than—independent cosmological estimates [11, 12]. This weakness is itself informative: the thermodynamic cost of self-modeling is negligible compared to the gravitational entropy budget, confirming that the entropy gradient theorem constrains the *qualitative existence* of an arrow of time, not its quantitative magnitude.

VII. DISCUSSION

A. What this paper establishes and what remains open

The central results of this paper are the entropy gradient theorem (Theorem 10), which constrains *all* self-modelers via three independent routes, and the structural convergence of the thermodynamic and algebraic chains on time-orientation, which narrows Gap C: complexification is necessary for Standard Model–like observers within $h_3(\mathbb{O})$, but the status of non-SM self-modelers in non-complexified blocks remains open.

Two independent arguments lead to time-orientation. The *thermodynamic chain* runs: self-modeling $\to$ Landauer bound $\to$ free energy $\to$ entropy gradient $\to$ time-orientation (Theorem 10). The *algebraic chain* runs: complexification $\to$ chirality $\to$ time-orientation (Theorem 7). The convergence of these chains on the same geometric structure—time-orientation—is a structural insight connecting observer thermodynamics to particle physics. It is not a single deductive chain in which one implies the other.

TABLE IV. Predictions of the self-modeling thermodynamics framework. "Strength" indicates the epistemic status: THEOREM (rigorous under stated assumptions), SELECTION (logical chain with identified weakest link), MATHEMATICAL (follows from definitions), ESTIMATE (order-of-magnitude, model-dependent), STRUCTURAL (qualitative connection without a quantitative formula). Assumptions A1–A7 are framework assumptions (Sec. V); A8–A10 are cosmological inputs (Sec. VI E).

#	Prediction
P1	Entropy gradient required: $S(t) < S_{\max}$ for $\rho > 0$
P2	$\Delta S \geq I(B; M)$ per update cycle
P3	$\rho = 0$ at thermal equilibrium
P4	Complexification necessary for SM-like observers (chirality requires Spin(10)); non-SM self-modelers in non-complexified blocks remain
P5	ρ peaks at $I = S_B/2$ with $\rho_{\max} = S_B/4$
P6	$\rho \to 0$ at heat death
P7	$\Delta S_{\text{Landauer}} \sim 10^{28}$ nats
P8	Landauer ~ 94 orders weaker than Penrose
P9	$\tau_{\text{exhaust}} \sim 10^{111}$ s $\gg t_U$

This convergence narrows Gap C but does not close it. For SM-like observers—those whose matter content is described by the chiral representation arising from Cl(6)—complexification of $V_{1/2}$ is necessary, because without it the chirality operator has no eigenstates and the L/R decomposition does not exist. For such observers, the algebraic chain is blocked. For the same observers, the thermodynamic chain independently requires time-orientation, which in the SM-like setting presupposes the geometric structures that complexification provides.

The nine-link chain (Table II) is accordingly updated: six links are proved algebraically (L1–L3, L5, L8, L9), one is updated with a geometric consequence (L7), one is narrowed via selection for SM-like observers (L6, this paper), and one remains assumed (L4, Gap B1).

We emphasize what remains open. Four gaps are not addressed here:

- **Gap A** ($h_3(\mathbb{O})$ identification): Why is the exceptional Jordan algebra $h_3(\mathbb{O})$ the "universe algebra"? Paper 5's non-composability argument [7] motivates this choice but does not uniquely determine it.

- **Gap B1** (rank-1 idempotent): What selects the observer's idempotent E_{11}? All rank-1 idempotents are F_4-conjugate, so any selection principle must invoke additional structure beyond the algebra's automorphism group.

- **Gap B2** (complex structure u): What selects the specific $u \in S^6$? The moduli space S^6 is six-dimensional; the spectral action might generate a potential with a preferred minimum.

- **Gap SA** (spectral action): The algebraic data established by Papers 5–8—gauge group, chiral representation, and complexified Peirce space—are the *input* to the spectral action principle [13], which should produce the full GR + SM Lagrangian as *output*. This computation has not been performed for the $h_3(\mathbb{O})$-based spectral triple.

The selection narrowing of Gap C is explicitly weaker than an algebraic proof would be. An algebraic proof would close Gap C unconditionally: $V_{1/2}$ *must* complexify as a consequence of the algebra's structure. The selection argument narrows it conditionally: among SM-like blocks with $\rho > 0$, $V_{1/2}$ is complexified. This is a *necessary condition* for SM-like particle physics (complexification is required for chirality), not a *sufficient condition* (complexification alone does not guarantee a viable self-modeler).

a. What this paper does not claim. We state four non-claims explicitly.

1. This paper does *not* prove that $\rho = 0$ for all non-complexified blocks. The entropy gradient theorem applies to all self-modelers regardless of their matter content; a non-complexified block could in principle support a non-SM self-modeler that satisfies the thermodynamic chain without chiral fermions.

2. This paper does *not* prove that all experience requires complexification. The experiential measure ρ depends on the self-modeling axioms (L4, Gap B1), and self-modelers with non-SM physics might exist under those axioms.

3. Non-SM self-modelers—if they exist—are inaccessible to us: our observations are mediated by the SM-like physics that the algebraic chain describes. They are not forbidden, merely outside the scope of this framework's predictions.

4. The program explains *our* physics, not all possible physics. This has the same epistemic status as anthropic arguments for spatial dimensionality: the framework constrains the physics *compatible with observers like us*, not the totality of possible physics.

B. Selection vs. algebraic derivation

The epistemic status of the Gap C narrowing deserves careful statement. The experiential measure ρ selects complexified blocks among SM-like observers, but does not force complexification algebraically. The algebraic obstruction—the $V_1 = \mathbb{R} \cdot E_{11}$ bottleneck in the Peirce decomposition [5]—remains genuine. No algebraic mechanism within the Jordan structure of $h_3(\mathbb{O})$ promotes the real $V_{1/2} = \mathbb{O}^2$ to the complex $V_{1/2}^{\mathbb{C}} = S_{10}^{+}$. What the two-chain argument provides is a *structural* filter for SM-like observers: the algebraic chain shows that such observers require complexification for chirality, and the thermodynamic chain independently shows that all self-modelers require time-orientation. The convergence of both chains on the same geometric prerequisite makes the case that SM-like self-modelers inhabit complexified blocks. Whether non-complexified blocks can support non-SM self-modeling is a separate question that this paper leaves open.

An analogy clarifies the logical structure. In evolutionary biology, natural selection does not force specific mutations to occur; it acts on the phenotypic consequences of mutations that have already occurred. Similarly, the experiential measure does not force structure space to contain complexified blocks; it acts on the thermodynamic consequences of blocks that are already present. The selection is over *existing* structure, not a dynamical process that creates new structure.

This connects to Hoffman's interface theory of perception [16]. In the self-modeling framework, a block with $\rho = 0$ is a passive interface: it exists in structure space but supports no self-modeling activity. A block with $\rho > 0$ is an active self-model: the body–model correlations are maintained at thermodynamic cost, and the experiential measure registers positive weight. The distinction between passive and active is thermodynamic, not fitness-theoretic: what separates $\rho > 0$ from $\rho = 0$ is the availability of an entropy gradient, not an external fitness function. For SM-like observers, the additional algebraic requirement of chirality ties the thermodynamic condition to complexification specifically. For hypothetical non-SM self-modelers, the thermodynamic condition still applies but the algebraic link to complexification does not.

C. The chirality–thermodynamics nexus

Corollary 14 identifies time-orientation as the common geometric prerequisite for both the chirality of the Standard Model and the thermodynamic arrow of time. This is a structural connection, not a coincidence.

The chirality side of the connection is algebraic and geometric: the choice $u \in S^6$ determines the internal chirality via the Cl(6) volume form ω_6 [6], and its physical realization as spacetime Weyl spinors requires a globally consistent sign for the chirality operator Γ_*, which in turn requires time-orientation (Proposition 5, Theorem 7). The thermodynamic side is dynamical: the entropy gradient theorem (Theorem 10) shows that self-modelers require $S(t) < S_{\max}$, defining a preferred temporal direction (the "future" is the direction of entropy increase), which on a Lorentzian manifold is a time-orientation.

Both requirements trace back to self-modeling. The algebraic choice u is made within the $h_3(\mathbb{O})$ framework that the self-modeling axioms motivate [6, 7]. The entropy gradient arises from the thermodynamic cost of self-modeling (Theorem 2). Time-orientation is the geometric structure that both consequences demand, and it is this demand—not temporal coincidence—that links particle physics chirality to the arrow of time.

D. CP violation: a structural remark

The chirality–time-orientation link established in Sec. IV admits a qualitative extension to CP violation. The CPT theorem (valid in any local Lorentz-invariant quantum field theory) implies that CP violation entails T violation. T violation—the dynamical asymmetry between forward and backward temporal evolution—can only be defined on a time-oriented spacetime, since "forward" and "backward" presuppose a preferred temporal direction. The resulting structural triangle is:

1. Chirality $\leftrightarrow$ time-orientation (Theorem 7);
2. Self-modeling $\rightarrow$ entropy gradient $\rightarrow$ time-orientation (Theorem 10);
3. CP violation $\rightarrow$ T violation $\rightarrow$ time-orientation (CPT theorem).

All three arrows converge on time-orientation as a shared geometric prerequisite. This suggests that chirality, the thermodynamic arrow of time, and CP violation are structurally connected through their common dependence on the same spacetime structure.

We state explicitly what this structural remark does *not* claim. The framework does not derive the magnitude of CP violation: the Jarlskog invariant, the CKM phase, and the strong CP parameter are Standard Model quantities determined by Yukawa couplings, which lie outside the scope of the self-modeling axioms. The framework does not derive baryon asymmetry from the entropy gradient: Sakharov's conditions (a) (baryon number violation) and (b) (CP violation) are independent requirements that the entropy gradient theorem does not address. And no equation in this framework relates the entropy gradient rate to the CP violation magnitude. The connection is structural and qualitative: all three phenomena require time-orientation, and the self-modeling framework shows why time-orientation is necessary.

E. Assumption hierarchy

The master theorem (Theorem 13) rests on seven assumptions (Table III), and the quantitative predictions of Sec. VI require three additional cosmological inputs (A8–A10). The assumptions are not equally strong; their failure modes differ qualitatively.

Assumption A3 (closed system equilibration) is the strongest. It connects local thermodynamics—the Landauer cost per self-modeling cycle—to a cosmological conclusion—the necessity of a low-entropy past. If A3 fails because an infinite free energy source exists (e.g., an infinite-dimensional Hilbert space, or unknown physics that provides work without depleting a reservoir), the chain breaks at Link 3 of Proposition 3: the entropy gradient theorem still holds via Routes B and C, but Route A's quantitative connection to the Landauer cost is weakened.

Assumptions A6–A7 (continuum limit and Lorentzian signature) are standard in the sense that they are shared by essentially all approaches to emergent spacetime, but they remain unvalidated from first principles within the self-modeling framework. Paper 6's lattice [17] provides the microscopic structure; the claim that its continuum limit yields a smooth Lorentzian manifold is physically motivated by Lieb-Robinson bounds but not rigorously proved. If A6 or A7 fail, the three-consequence theorem (Theorem 7) reduces to Paper 7's two-consequence theorem: consequences (a) and (b) (gauge group and chirality) survive as purely algebraic results, while consequence (c) (time-orientation requirement) and the complexification selection argument are lost.

Assumptions A8–A10 (cosmological entropy budget, Penrose's initial entropy estimate, and self-modeling parameters) are external inputs to the framework, not consequences of it. They enter only the order-of-magnitude estimates P7–P9 of Sec. VI and do not affect the framework predictions P1–P6. Their uncertainty is large (several orders of magnitude in each parameter), but the qualitative conclusions are robust: the 94-order-of-magnitude gap between the Landauer deficit and the Penrose deficit (Eq. (49)) survives any plausible variation of these inputs.

F. Future directions

Five concrete extensions of the present work are identified, each connected to a specific open problem.

1. Spectral action computation (Gap SA). The algebraic data established by Papers 5–8 provide the input to the spectral action principle [13]: the gauge group $\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, the chiral representation from Cl(6), and the complexified Peirce space S_{10}^{+}. Computing the spectral action for an $h_3(\mathbb{O})$-based spectral triple would yield the full GR + SM Lagrangian, including the Higgs sector, Yukawa couplings, and the cosmological constant. The principal challenge is the non-associativity of the octonions, which prevents a straightforward application of the Connes–Chamseddine–Marcolli formalism [13].

2. Sharper Landauer bound from detailed self-modeling models. The Landauer bound of Theorem 2 gives the minimum thermodynamic cost of self-modeling per update cycle. This bound is tight for quasi-static erasure protocols but may be far from tight for realistic self-modeling dynamics. A detailed model of the self-modeling cycle—specifying the measurement, erasure, and writing steps concretely—would yield a tighter bound, potentially shifting the 94-order-of-magnitude gap of Eq. (49).

3. Generation structure. The present framework produces one generation of chiral fermions in the **16** of Spin(10). The Standard Model requires three generations. The "3" in $h_3(\mathbb{O})$ (the algebra of 3×3 matrices) is suggestive but no mechanism producing three copies of the **16** has been established. Furey [3] proposed a route via the full octonion algebra acting on Cl(6); Boyle [4] explored triality. Making any of these connections rigorous within the self-modeling framework is an open challenge.

4. Quantitative CP–entropy connection. The structural triangle of Sec. VII D identifies a qualitative link between CP violation and the entropy gradient. A quantitative version would require showing that gauge groups admitting CP-violating phases are selected for by the experiential measure—an argument analogous to the Gap C selection but operating at the level of the CKM structure rather than the complexification step. This lies well beyond the scope of the present work.

5. Experimental and observational signatures. The predictions of Sec. VI are primarily consistency checks rather than novel observational predictions. The most direct path to testable consequences is through the spectral action computation (direction 1), which would predict specific relations among Standard Model parameters. A less direct but potentially more distinctive signature is the exhaustion timescale τ_{exhaust}: any observation constraining the long-term thermodynamic fate of the universe constrains the self-modeling capacity.

G. Concluding remarks

The question "why complex numbers in quantum mechanics?" has been posed since the formalism's inception. Within the self-modeling framework, this paper provides a partial answer for SM-like observers: because chirality requires complexification, and chirality requires the same time-orientation that self-modeling independently demands.

Two independent chains converge on time-orientation. The thermodynamic chain: self-modeling demands an

entropy gradient (Theorem 10), which defines a time-orientation on the emergent spacetime. The algebraic chain: the choice $u \in S^6$ determines the gauge group and chirality (Theorem 7), and the physical realization of chirality as spacetime Weyl spinors requires time-orientation (Proposition 5). The complexification of $V_{1/2}$ enters through the algebraic chain: without it, the L/R decomposition from Cl(6) does not exist, and the chirality that requires time-orientation is absent.

The complexification step that seemed algebraically unmotivated in Paper 7 [6] is thereby given a physical rationale: it is necessary for SM-like observers, whose chiral matter content requires the geometric structure that self-modeling thermodynamics independently demands. This is not logical closure—non-SM self-modelers in non-complexified blocks are not excluded—but it is structural convergence: two independent arguments, one thermodynamic and one algebraic, identify time-orientation as indispensable, and for SM-like physics, time-orientation requires complexification.

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Appendix A: Detailed Derivations

This appendix collects derivations supporting the main text that would interrupt the narrative flow if included in full. Section A 1 gives the Kraus operator decomposition of the SWAP channel (4). Section A 2 develops the Sagawa–Ueda mapping referenced in Sec. III C. Section A 3 derives the general ρ_M channel formula (7).

1. Kraus operator derivation for the SWAP channel

The depolarizing channel (4),

$$\mathcal{E}(\rho_B) = (1-p)\,\rho_B + p\,\frac{\mathbb{1}}{2}\,, \qquad p = \sin^2(Jt)\,, \tag{A1}$$

admits an explicit Kraus decomposition. We rewrite the maximally mixed contribution using the Pauli identity

$$\sum_{j=1}^{3} \sigma_j\,\rho\,\sigma_j = 2\,\mathrm{Tr}(\rho)\,\mathbb{1} - \rho\,, \tag{A2}$$

valid for any 2×2 matrix ρ, which gives $\mathbb{1}/2 = \big(\rho + \sum_j \sigma_j \rho \sigma_j\big)/4$ when $\mathrm{Tr}(\rho) = 1$. Substituting into (A1):

$$\begin{aligned}\mathcal{E}(\rho_B) &= (1-p)\,\rho_B + \frac{p}{4}\Big(\rho_B + \sum_{j=1}^{3}\sigma_j\,\rho_B\,\sigma_j\Big)\\ &= \Big(1-\tfrac{3p}{4}\Big)\,\rho_B + \frac{p}{4}\sum_{j=1}^{3}\sigma_j\,\rho_B\,\sigma_j\,.\end{aligned} \tag{A3}$$

This is a sum of four terms $K_i\,\rho_B\,K_i^\dagger$ with the Kraus operators

$$K_0 = \sqrt{1-\tfrac{3p}{4}}\;\mathbb{1}\,, \qquad K_j = \frac{\sqrt{p}}{2}\,\sigma_j \quad (j=1,2,3)\,. \tag{A4}$$

a. Trace preservation. Using $\sigma_j^\dagger = \sigma_j$ and $\sigma_j^2 = \mathbb{1}$:

$$\begin{aligned}\sum_{i=0}^{3} K_i^\dagger K_i &= \Big(1-\tfrac{3p}{4}\Big)\,\mathbb{1} + \frac{p}{4}\sum_{j=1}^{3}\sigma_j^2\\ &= \Big(1-\tfrac{3p}{4}\Big)\,\mathbb{1} + \frac{3p}{4}\,\mathbb{1} = \mathbb{1}\,.\end{aligned} \tag{A5}$$

The channel is trace-preserving for all $p \in [0,1]$.

b. Unitality. Acting on $\mathbb{1}/2$:

$$\begin{aligned}\mathcal{E}\Big(\frac{\mathbb{1}}{2}\Big) &= \Big(1-\tfrac{3p}{4}\Big)\frac{\mathbb{1}}{2} + \frac{p}{4}\sum_{j=1}^{3}\sigma_j\frac{\mathbb{1}}{2}\sigma_j\\ &= \Big(1-\tfrac{3p}{4}\Big)\frac{\mathbb{1}}{2} + \frac{3p}{4}\,\frac{\mathbb{1}}{2} = \frac{\mathbb{1}}{2}\,,\end{aligned} \tag{A6}$$

where we used $\sigma_j(\mathbb{1}/2)\sigma_j = \mathbb{1}/2$. The channel is unital, confirming Eq. (6) in the main text.

c. Complete positivity. The Kraus operators (A4) are well-defined (i.e., the prefactors are real) when $1 - 3p/4 \geq 0$, i.e., $p \leq 4/3$. Since $p = \sin^2(Jt) \leq 1 < 4/3$, this condition is always satisfied. The Kraus representation manifestly defines a CP map [18].

2. Sagawa–Ueda mapping for the coherence loophole

Section III C (argument R3) invoked the generalized Jarzynski equality of Sagawa and Ueda [19] as an independent cross-check on the Landauer bound (Theorem 2). Here we develop the mapping in detail and verify consistency.

a. Generalized Jarzynski equality. For a system undergoing measurement followed by feedback control [19], the work W performed on the system satisfies

$$\big\langle e^{-\beta(W-\Delta F)}\big\rangle = e^{I_{fc}}\,, \tag{A7}$$

where ΔF is the equilibrium free energy change and I_{fc} is the efficacy of feedback control (transfer entropy). By Jensen's inequality ($\langle e^X\rangle \geq e^{\langle X\rangle}$), the mean work satisfies

$$\langle W\rangle \geq \Delta F - k_{\mathrm{B}}T\,I_{fc}\,. \tag{A8}$$

For a cyclic process ($\Delta F = 0$), the dissipated work is bounded below by $k_{\mathrm{B}}T\,I_{fc}$.

b. Mapping to the self-modeling cycle. The self-modeling cycle maps onto the Sagawa–Ueda framework as:

Self-modeling	Sagawa–Ueda
Body B	System
Model M	Memory / controller
Lüders product (testing B)	Measurement
Model update	Feedback control
$I(B;M)$	Mutual information

In this mapping, the body plays the role of the system to be controlled, the model plays the role of the measurement memory, the Lüders product [7] implements the measurement step, and the subsequent model update implements feedback.

c. Consistency with Theorem 2. The two bounds constrain the same physical quantity (work per cycle) from different starting points:

- **Landauer route** (Theorem 2): the erasure step of the Bennett decomposition requires $W \geq k_{\mathrm{B}}T\, I(B;M)$.

- **Sagawa–Ueda route**: for a cyclic process, $W \geq k_{\mathrm{B}}T\, I_{fc}$. In the pure-erasure limit ($I_{fc} \to 0$, no feedback advantage), this reduces to the standard Landauer bound. When $I_{fc} > 0$, the feedback step can extract at most $k_{\mathrm{B}}T\, I_{fc}$, but the memory reset still costs $k_{\mathrm{B}}T\, S_{\mathrm{vN}}(\rho_M) \geq k_{\mathrm{B}}T\, I(B;M)$ by the Reeb–Wolf bound [14]. The full-cycle cost is therefore consistent with Eq. (17).

d. Numerical verification. We tested 50 random two-qubit states and confirmed that both bounds give non-negative work costs in every case. For pure bipartite states $|\psi\rangle_{BM}$, the relation $I(B;M) = 2\, S_{\mathrm{vN}}(\rho_M)$ holds exactly (verified to numerical precision $< 10^{-6}$), confirming that the cycle cost exceeds the single-register erasure cost by the expected factor of two.

e. Epistemic status. The mapping from the self-modeling cycle to the Sagawa–Ueda feedback framework is physically motivated but not axiomatically derived from the Paper 5 axioms [7]. It serves as an independent consistency check: two different thermodynamic frameworks, applied to the same physical cycle, yield compatible bounds. The Landauer bound (Theorem 2) stands on its own; the Sagawa–Ueda mapping strengthens confidence in its robustness.

3. General ρ_M channel formula

The main text (Eq. (7)) states the reduced dynamics for an arbitrary model state ρ_M. Here we derive the result from the unitary evolution (2) by explicit partial trace.

Starting from $\rho_{BM}(t) = U(t)\,(\rho_B \otimes \rho_M)\,U(t)^\dagger$ with $U(t) = \cos(Jt)\,\mathbb{1} - i\sin(Jt)\,F$, the reduced state $\rho_B(t) = \mathrm{Tr}_M[\rho_{BM}(t)]$ expands into four terms:

$$\begin{aligned}\rho_B(t) = \cos^2(Jt)\ \mathrm{Tr}_M[\rho_B \otimes \rho_M] \\ + \sin^2(Jt)\ \mathrm{Tr}_M[F(\rho_B \otimes \rho_M)F] \\ - i\sin(Jt)\cos(Jt)\ \mathrm{Tr}_M[F(\rho_B \otimes \rho_M)] \\ + i\sin(Jt)\cos(Jt)\ \mathrm{Tr}_M[(\rho_B \otimes \rho_M)F]\,. \end{aligned} \tag{A9}$$

We evaluate each term using the index structure of the SWAP operator, $F_{ac,bd} = \delta_{ad}\delta_{cb}$, where (a,c) and (b,d) label the B and M indices respectively.

a. Term 1. $\mathrm{Tr}_M[\rho_B \otimes \rho_M] = \rho_B\ \mathrm{Tr}(\rho_M) = \rho_B$.

b. Term 2. Since $F(\rho_B \otimes \rho_M)F = \rho_M \otimes \rho_B$ (SWAP exchanges the two tensor factors), $\mathrm{Tr}_M[\rho_M \otimes \rho_B] = \rho_M\ \mathrm{Tr}(\rho_B) = \rho_M$.

c. Term 3. Using the index form of F:

$$\big[F(\rho_B \otimes \rho_M)\big]_{ac,bd} = (\rho_B)_{cb}\,(\rho_M)_{ad}\,. \tag{A10}$$

Tracing over M (setting $d = c$ and summing):

$$\big[\mathrm{Tr}_M\big(F(\rho_B \otimes \rho_M)\big)\big]_{ab} = \sum_c (\rho_M)_{ac}\,(\rho_B)_{cb} = (\rho_M\,\rho_B)_{ab}\,. \tag{A11}$$

d. Term 4. By an analogous computation:

$$\big[(\rho_B \otimes \rho_M)F\big]_{ac,bd} = (\rho_B)_{ad}\,(\rho_M)_{cb}\,, \tag{A12}$$

so that

$$\big[\mathrm{Tr}_M\big((\rho_B \otimes \rho_M)F\big)\big]_{ab} = \sum_c (\rho_B)_{ac}\,(\rho_M)_{cb} = (\rho_B\,\rho_M)_{ab}\,. \tag{A13}$$

e. Combining. Substituting terms 1–4 into Eq. (A9):

$$\rho_B(t) = \cos^2(Jt)\ \rho_B + \sin^2(Jt)\ \rho_M - i\sin(Jt)\cos(Jt)\ \big(\rho_M\rho_B - \rho_B\rho_M\big)\,. \tag{A14}$$

Recognizing the commutator $[\rho_M, \rho_B] = \rho_M\rho_B - \rho_B\rho_M$, this reproduces Eq. (7):

$$\boxed{\rho_B(t) = \cos^2(Jt)\ \rho_B + \sin^2(Jt)\ \rho_M - i\sin(Jt)\cos(Jt)\ [\rho_M, \rho_B]} \tag{A15}$$

f. Consistency checks.

1. $\rho_M = \mathbb{1}/2$*:* The commutator $[\mathbb{1}/2, \rho_B] = 0$ for all ρ_B, recovering the depolarizing channel (3).

2. $[\rho_M, \rho_B] = 0$*:* When the model and body states commute, the cross term vanishes and $\rho_B(t) = \cos^2(Jt)\ \rho_B + \sin^2(Jt)\ \rho_M$, a classical mixing of the two states.

3. *Non-unitality:* $\mathcal{E}(\mathbb{1}/2) = \cos^2(Jt)\,(\mathbb{1}/2) + \sin^2(Jt)\,\rho_M \neq \mathbb{1}/2$ unless $\rho_M = \mathbb{1}/2$. The channel is unital only for the maximally mixed model.

4. *Entropy decrease:* For $\rho_M \neq \mathbb{1}/2$, the non-unital channel can decrease the body's entropy. Numerical verification with $\rho_M = \mathrm{diag}(0.9, 0.1)$ shows $\Delta S < 0$ for 56% of state–time points in a parametric sweep (9900 samples), as reported in Sec. II A.

Appendix B: Assumption Register

The main text (Table III) summarizes the seven framework assumptions entering the master theorem (Theorem 13). This appendix provides the complete register, organized into two tiers: *framework assumptions* (A1–A7), which define the theoretical setting and are common to all results, and *model-dependent parameters* (A8–A10), which enter only the order-of-magnitude estimates of Sec. VI. For each assumption we record its source, severity, current status, what evidence would close or validate it, and which results depend on it.

1. Framework assumptions

Table V expands on the main-text summary. Severity levels are: STANDARD (widely accepted, no serious prospect of failure), MODEL CHOICE (replaceable without changing qualitative conclusions), MEDIUM (plausible but unproven), and HIGH (strongest assumption in the chain; failure would weaken key conclusions).

Assumption A3 is the strongest in the chain. If it fails—e.g., if recurrence times for self-modeling Hamiltonians are shorter than expected—then Route A of the entropy gradient theorem (Sec. V A) is weakened, though Routes B and C survive. Assumptions A6 and A7 are needed only for the SM-observer complexification requirement [part (II) of the master theorem]; the entropy gradient [part (I)] is independent of both. A5 (experiential density) enters via Route C and part (II) step (d).

2. Model-dependent parameters

The order-of-magnitude estimates P7–P9 of Sec. VI require three cosmological or biological inputs external to the framework. Table VI records each parameter, its fiducial value, plausible range, and sensitivity.

Varying all three parameters simultaneously across their full plausible ranges shifts $\Delta S_{\text{Landauer}}$ by at most ~ 6 orders of magnitude (from $\sim 10^{25}$ to $\sim 10^{31}$ nats). The 94-order gap with the Penrose deficit (Eq. (49)) narrows to at worst ~ 88 orders of magnitude: the qualitative conclusion is robust.

3. Dependency map

Table VII records which assumptions are required by each theorem, proposition, and named result in the paper. An entry "+A3" indicates that the assumption strengthens the result (e.g., extends it from a single-cycle bound to a cosmological statement) but is not required for the core claim.

The dependency structure has two notable features. First, the entropy gradient theorem has three independent routes with different assumption subsets (Table I), so no single assumption failure destroys the conclusion. Second, the model-dependent parameters A8–A10 enter only the quantitative estimates; all structural results (Landauer bound, chain theorem, Gap C resolution) are independent of cosmological inputs.

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TABLE V. Full framework assumption register (A1–A7). "What would close it" describes the minimal result that would either validate the assumption or render it unnecessary. "Depends" lists results that rely on each assumption.

ID	Assumption	Source	Severity	Status	What would close it
A1	Finite-dimensional quantum mechanics	Paper 5 [7]	STANDARD	Accepted	—
A2	Thermal contact at temperature T	Standard thermodynamics	STANDARD	Accepted	—
A3	Closed/semi-closed system equilibrates on finite timescales	Statistical mechanics	HIGH	Open	Rigorous equilibration bound for self-modeling Hamiltonians (e.g., extending [20] to SWAP lattices)
A4	SWAP lattice dynamics ($H = JF$)	Paper 6 [17]	MODEL CHOICE	Accepted	Show qualitative results hold for general nearest-neighbor H
A5	Experiential density formula: $\rho = I(B;M)(1 - I(B;M)/S_B)$	Paper 5 [7]	STANDARD	Accepted	— (definitional within Paper 5 axioms)
A6	Continuum limit of self-modeling lattice yields smooth Lorentzian manifold	Paper 6 [17]	MEDIUM	Unvalidated	Rigorous continuum-limit construction (cf. causal-set–manifold correspondence)
A7	Emergent spacetime has Lorentzian signature	Paper 6 causal structure [17]	MEDIUM	Unvalidated	Derive signature from causal axioms (e.g., via Malament's theorem [21])

TABLE VI. Model-dependent parameters (A8–A10). These enter only the order-of-magnitude predictions P7–P9; the framework predictions P1–P6 are independent of all three. "Sensitivity" indicates how the Landauer deficit $\Delta S_{\text{Landauer}}$ or the exhaustion timescale τ_{exhaust} scales with each parameter.

ID	Parameter	Fiducial value	Plausible range	Sensitivity
A8	Cosmological entropy budget $S_{\max}$	$\sim 10^{122}$ nats (Bekenstein)	10^{120}–10^{124}	$\tau_{\text{exhaust}} \propto S_{\max}$; linear in ΔS
A9	Neural self-modeling dimension d_M	$\sim 10^{11}$	10^{9}–10^{13}	Logarithmic in $I(B;M)$; linear in $\Delta S_{\text{Landauer}}$
A10	Self-modeling update rate ν	~ 10 Hz	1–100 Hz	Linear in N; linear in τ_{exhaust}

TABLE VII. Dependency map: assumptions required by each result. Parenthesized entries are needed only for specific subresults (e.g., a particular route or corollary).

Result	Assumptions
Landauer bound (Thm. 2)	A1, A2
Coherence loophole closure	A1, A2 (+Markovian bath for R2)
Chain theorem (Prop. 3)	A1, A2, A3
Entropy gradient, Route A	A1, A2, A3
Entropy gradient, Route B	A1, A3
Entropy gradient, Route C	A1, A5
Three-consequence theorem (Thm. 7)	A6, A7
Gap C resolution (Thm. 11)	A1–A7
Master theorem (Thm. 13)	A1–A7
Quantitative estimates P7–P9	A1–A7 + A8–A10

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